

# Trend-Following in Financial Markets

From the P&L identity to a working system

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What this is. Trend-following is the oldest systematic trading strategy still in industrial use, and the one whose theory is most often stated badly. This document builds it from first principles: what the object is mathematically, what quantity a trend-follower is actually harvesting, how that quantity is estimated in practice, when and why it disappears, and what the risk apparatus around it is for.

How to read it. Sections 1–3 are conceptual and historical — the mental model, the candidate mechanisms, and how the industry arrived at its current shape. Section 4 is the bibliography. Sections 5–7 are the technical core: §5 derives what trend-following earns and what shape its payoff has, §6 surveys every serious way to estimate a trend, §7 shows that most of §6 is one estimator with three knobs. Sections 8–11 are engineering: portfolio construction, degradation, evaluation, and risk. Section 12 synthesises.

If you want the single most important idea and nothing else, read §5.4, §5.5 and §7.2. If you are implementing, read §8 and §11 and treat the rest as reference.

Relationship to the momentum note. A companion document, Momentum in Financial Markets, treats momentum — the statistical phenomenon of return predictability from past returns. This document treats trend-following — the trading system. They are not the same thing, and §1.3 makes the distinction precise. Where the momentum note goes deeper on signal measurement and cross-sectional design, this one goes deeper on the P&L functional, position sizing, and risk. Each stands alone; the overlap is deliberate and small.

Epistemic tags. Claims are flagged by status:

- [Fact] — replicated across independent datasets or implementations, with broad agreement among people who have looked.
- [Contested] — documented, but with live disagreement about magnitude, robustness, or cause.
- [Hypothesis] — a proposed mechanism, not decisively tested.
- [Practice] — practitioner convention. May well be right; the evidence is private or absent.

Untagged sentences are definitions, derivations, or arithmetic — things that are true by construction rather than by evidence. Several of the derivations below I verified numerically while writing; where a number in the text comes from my own simulation rather than from a paper, I say so.

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Notation.  $P_t$  is the price at time  $t$  and  $p_t = \ln P_t$  the log price;  $r_t = p_t - p_{t-1}$  is the one-bar log return. A bar is whatever sampling interval you work in, and  $A$  is the number of bars per year (252 or 256 for daily data).  $\mathcal{F}_t$  is the information set available at  $t$ .

$\pi_t$  is the position — the signed exposure held over the interval  $(t, t + 1]$ , chosen using  $\mathcal{F}_t$  only.  $s_t$  is a raw signal, and  $x_t$  specifically denotes the exponentially weighted trend estimate  $x_t = (1 - \alpha)x_{t-1} + \alpha r_t$  with decay  $\alpha \in (0, 1)$ , whose span is  $2/\alpha - 1$  bars.  $w_k$  is the weight a linear filter places on  $r_{t-k}$ ; the collection  $\{w_k\}$  is the kernel.  $g(\cdot)$  is the response function mapping a normalised signal to a position.

$\gamma_k = \text{Cov}(r_t, r_{t+k})$  is the lag- $k$  autocovariance and  $\rho_k = \gamma_k/\gamma_0$  the autocorrelation;  $\gamma_0 = \sigma^2$  is the per-bar return variance.  $\text{VR}(q)$  is the variance ratio at horizon  $q$ .  $\hat{\sigma}_t$  is an estimate, formed from  $\mathcal{F}_t$ , of per-bar return volatility, and  $\sigma^*$  is a target volatility.

$L$  denotes a lookback in bars and  $n$  a moving-average window; both are “how far back”, but  $L$  appears in return-space rules and  $n$  in price-space ones. In the hidden-trend model of §5.8,  $\mu_t$  is the unobserved drift,  $\phi$  its autoregressive coefficient,  $\tau = 1/(1 - \phi)$  its persistence timescale in bars,  $\sigma_\varepsilon$  the noise volatility,  $\theta = \text{sd}(\mu_t)/\sigma_\varepsilon$  the per-bar signal-to-noise ratio, and  $K$  the steady-state Kalman gain.

Across markets,  $N$  is the number of instruments and  $\bar{\rho}$  the average pairwise correlation of strategy returns — distinct from  $\rho_k$ , which is always an autocorrelation of one series. Two symbols carry a qualification:  $\alpha$  is always an EWMA decay constant, never a significance level, and  $\sigma$  is always a return volatility except where subscripted  $\sigma_\eta, \sigma_\varepsilon$  as model innovations.

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## 1. What trend-following is

### 1.1 The wrong intuition

Ask a practitioner what trend-following is and you will usually get some version of: “you identify that a market is trending, and you ride the trend.” This description is not wrong so much as it is unusable, because it presupposes that “is trending” is a property a market has, which you could in principle detect.

It isn’t. Consider what you actually observe: a finite sequence of prices. Every such sequence, without exception, decomposes into a smooth part and a rough part in infinitely many ways. Fit a 50-day moving average and there is a trend; fit a 5-day one and the same data has six trends; fit a straight line and there is one. None of these decompositions is more correct than the others, because “trend” is not a feature of the data — it is a feature of the model you impose on the data. A price path does not come with a trend attached any more than a photograph comes with a Fourier transform attached.

The second wrong intuition is the physical one, inherited from the word “momentum”: a market in motion tends to stay in motion. There is no conserved quantity in a price series, no mass, no inertia. If you start here you will build systems that assume persistence and be surprised by every reversal.

The third — and the most costly in practice — is that trend-following is a prediction problem. It looks like one: you estimate something from the past and take a position on the future. But the estimate a trend-follower forms is almost never accurate enough to be called a forecast in any ordinary sense. §5.9 makes this quantitative: the return predictability that supports the entire managed futures industry corresponds to a daily return autocorrelation of roughly 0.005, which cannot be distinguished from zero in a single market’s ten-year history. If you evaluate a trend signal as a forecast, you will conclude correctly that it is a terrible forecast, and incorrectly that it is worthless.

### 1.2 The right starting point: a map from paths to positions

Here is the definition to hold, and it is deliberately mechanical:

A trend-following system is a function that maps the observed price history to a position, with the property that the position is an increasing function of recent price changes.

$$\pi_t = F(p_t, p_{t-1}, p_{t-2}, \dots), \quad \frac{\partial \pi_t}{\partial r_{t-k}} \geq 0 \text{ for all } k \geq 0$$

That is the whole object. Note what the definition does not contain: no claim that a trend exists, no forecast, no latent state, no regime. It is a rule, and rules can be evaluated without believing anything about the data-generating process. The monotonicity condition — positions increase in past returns — is what makes it trend-following rather than mean-reversion, and the requirement that  $\pi_t$  use only information through  $t$  is what makes it implementable.

Everything interesting follows from asking two questions about this function:

1. What does it earn? The P&L is  $\sum_t \pi_t r_{t+1}$ , and because  $\pi_t$  is a function of past returns, the expected P&L is a functional of the return process’s autocovariance structure. §5 works this out exactly. The answer — trend-following is long autocovariance and long squared drift — is the organising fact of the field.
2. What shape is the payoff? Because  $\pi_t$  grows with past returns, the strategy is long when prices have risen and short when they have fallen, which means its P&L is a convex function of the price path. §5.5 and §5.6 make this precise,

and the convexity turns out to be exactly, algebraically, a long position in long-horizon variance against short-horizon variance.

### 1.3 The master form

Essentially every trend system in production is the same four-stage pipeline. Writing it down once saves enormous confusion later, because the “hundreds of indicators” in the practitioner literature are choices of one or two stages with everything else held fixed.

$$\pi_{i,t} = \underbrace{\frac{\sigma^*}{\hat{\sigma}_{i,t}}}_{(3) \text{ risk scaling}} \cdot \underbrace{g\left(\underbrace{\sum_{k \geq 0} w_k \tilde{r}_{i,t-k}}_{(1) \text{ kernel}}\right)}_{(2) \text{ response}} \cdot \underbrace{c_t}_{(4) \text{ aggregation}}$$

where  $\tilde{r}_{i,t}$  is the (possibly normalised) return of market  $i$  at  $t$ , and  $c_t$  is a portfolio-level multiplier. The four stages:

| Stage                                          | What it does                                            | What varies between systems                                |
|------------------------------------------------|---------------------------------------------------------|------------------------------------------------------------|
| (1) Kernel $\{w_k\}$                           | Summarises the recent past into one number              | Shape and timescale of the weights on past returns         |
| (2) Response $g(\cdot)$                        | Converts the signal into a desired exposure             | Linear, capped, sign, or bump-shaped                       |
| (3) Risk scaling $\sigma^*/\hat{\sigma}_{i,t}$ | Converts exposure into a position of known risk         | Volatility estimator and target                            |
| (4) Aggregation $c_t$                          | Combines markets into a portfolio at a total risk level | Correlation handling, capital constraint, drawdown control |

A 200-day moving-average crossover, a Donchian channel breakout, a 12-month time-series momentum sign, and a Kalman-filtered local-linear-trend model are all points in this space. §6 enumerates them; §7 shows the equivalences, several of which are exact identities rather than analogies.

The stage that matters most is not the one practitioners argue about. §7.3 and §8.1 make the case that stage (3) dominates, stage (4) is second, and stages (1) and (2) — the “indicator”, the thing with a name and a fan club — matter least. In a simulation with realistic volatility clustering, moving from a raw signal to a volatility-scaled one took the Sharpe ratio from 0.04 to 0.36; changing the kernel’s timescale by a factor of two, from its optimum, cost about 5% of the Sharpe.

### 1.4 What trend-following is not

These distinctions cause real losses. The middle column is what each object actually is, formally; the right column is why it gets confused with trend-following.

| Concept                      | Formal object                                                                            | Why it is confused with trend-following                                                                                                                   |
|------------------------------|------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|
| Trend-following              | A map $\pi_t = F(\mathcal{F}_t)$ increasing in past returns                              | —                                                                                                                                                         |
| Momentum                     | The statistical claim $\mathbb{E}[r_{t+1} \mid r_{t-L:t}]$ is increasing in past returns | Trend rules monetise momentum, but they also monetise drift and convexity. A profitable trend backtest is not evidence of momentum                        |
| Drift                        | The unconditional mean $\mu = \mathbb{E}[r_t]$                                           | A long-only asset with positive drift produces a positive trend backtest with zero predictability (§5.3)                                                  |
| Trend, as a decomposition    | The low-frequency component $\tau_t$ in $p_t = \tau_t + c_t + \varepsilon_t$             | This is a modelling choice, not an observable. A perfectly trending line plus noise has negative return autocorrelation                                   |
| Technical analysis           | A heterogeneous body of chart-pattern heuristics                                         | Trend rules were born inside it, and share vocabulary. But a trend rule is a measurable function with a testable P&L; a head-and-shoulders pattern is not |
| Market timing                | Switching between an asset and cash on a signal                                          | A long-only trend rule is market timing, but trend-following in general is long/short and multi-market                                                    |
| Managed futures / CTA        | An industry and a fee structure                                                          | Most CTAs are trend-followers, so index returns are used as a proxy for the strategy. The proxy carries fees, survivorship, and strategy drift            |
| Buy-the-dip / mean reversion | $\partial\pi_t/\partial r_{t-k} < 0$                                                     | Literally the same functional with the opposite sign. The sign that is correct depends entirely on the horizon (§9.2)                                     |

Two of these deserve expansion, because they are the ones that corrupt research.

Trend is not momentum, and a backtest cannot tell them apart

Take a price process that is a deterministic upward line plus independent noise:

$$p_t = \mu t + \varepsilon_t, \quad \varepsilon_t \sim \text{iid}(0, \sigma_\varepsilon^2)$$

This series has a perfect trend in the decomposition sense — the low-frequency component is exactly  $\mu t$ . Its returns are  $r_t = \mu + \varepsilon_t - \varepsilon_{t-1}$ , an MA(1) with autocorrelation  $\rho_1 = -1/2$  and  $\rho_k = 0$  for  $k \geq 2$ . So the series has a perfect trend and anti-momentum: after an up move you should expect a down move.

Meanwhile  $r_t = \phi r_{t-1} + u_t$  with  $\phi > 0$  and zero mean has momentum and no trend at all — the level wanders with no low-frequency component.

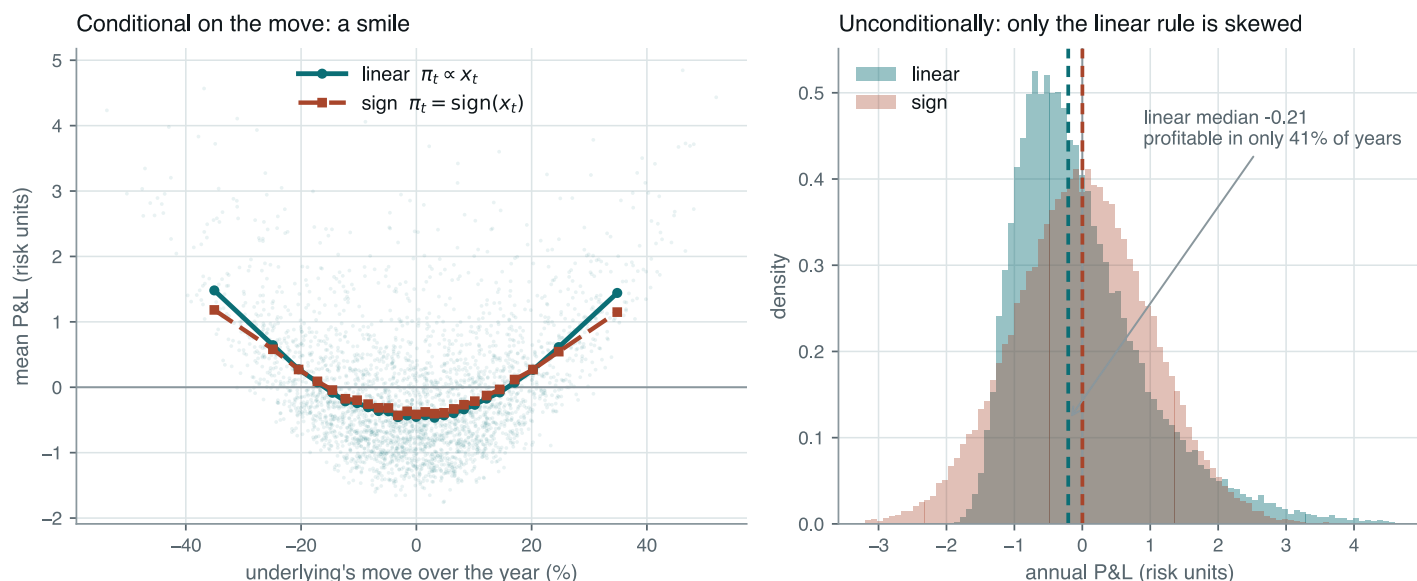
Trend is a property of the level; momentum is a property of the returns. A trend-following rule is indifferent: it makes money from drift, from trend, and from return autocorrelation, and its P&L does not tell you which. That is fine for earning money and fatal for research, because the three have completely different stability properties. Drift is stable and low-information; autocorrelation is fragile and high-information; and a rule that is secretly harvesting drift will fail the moment you apply it to a market without one.

The convexity is not a free option

Trend-following's payoff resembles a long straddle (§5.6), and this observation is often deployed as though it were good news — “trend-following gives you convexity for free”. It does not. Convexity is never free; the only question is how you pay for it. A straddle buyer pays a premium up front. A trend-follower pays continuously, in the whipsaw losses incurred when a signal reverses, and §5.5 shows the payment appearing explicitly as a  $-\sum_t r_t^2$  term in an exact algebraic identity. The strategy is profitable precisely when what it collects from trend persistence exceeds what it pays in realised variance, and not otherwise.

## 1.5 The payoff in one picture

The clearest way to see what kind of object this is: run a trend rule on paths with no predictability whatsoever — pure random walks — and plot what it earns.



Three things to take from it, all from my own simulation of 60,000 independent random-walk years with a 65-bar-span EWMA rule:

1. Conditional on the size of the year's move, the payoff is a smile. Big moves in either direction pay; small moves cost. This is the straddle analogy, and it holds for both response functions.
2. The mean is zero, as it must be. The underlying has no predictability; the simulated mean P&L was  $+0.003$  risk units against a standard deviation of 1.0. Nothing here is edge.
3. But the linear rule's P&L is right-skewed with a negative median. It was profitable in only 41% of years, with a median of  $-0.21$  risk units and a skew of  $+1.40$ . The sign-response rule, which discards the magnitude of the signal, was profitable in exactly 50% of years with skew  $-0.01$ .

Point 3 is the one people get wrong. A low hit rate is not evidence of a bad strategy, and a high hit rate is not evidence of a good one — with zero edge you can manufacture either, purely by choosing the response function. Potters and Bouchaud (2006) derived this analytically: for a trend rule on a driftless random walk the average gain per trade is exactly zero while the fraction of winning trades falls below one half and keeps falling as volatility rises. If you are diagnosing a trend system, the hit rate tells you about the shape of  $g$ , not about whether the system works.

### §1 Key takeaways

1. A market does not "have" a trend. A trend is a property of the model you impose, so "detecting a trend" is always a modelling choice with a Type I / Type II trade-off, never a measurement.
2. Define the object mechanically: a trend-following system is a map from price history to position, increasing in recent returns. Everything else is derived from that map.
3. The system is not a forecaster. The predictability it exploits is far too small to detect in one market (§5.9) and is only visible in aggregate.
4. Every production system is the same four-stage pipeline — kernel, response, risk scaling, aggregation. Named indicators are choices of the first two.
5. Risk scaling matters more than the indicator. In simulation, adding volatility normalisation moved the Sharpe from 0.04 to 0.36; halving or doubling the kernel timescale cost about 5%.
6. Trend  $\neq$  momentum  $\neq$  drift. A profitable trend backtest is consistent with zero return predictability, so it is not evidence of momentum.
7. The payoff is option-like, and the option is paid for, continuously, in realised variance. Convexity is a cost structure, not a gift.
8. Hit rate is a property of the response function, not of the edge. On a pure random walk a linear trend rule wins 41% of years and a sign rule wins 50%, with identical (zero) expected return.

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## 2. Why trends could exist

### 2.1 The objection that has to be answered first

If prices reflected all available information, past prices would be useless for predicting future ones, and every rule of the form  $\pi_t = F(\mathcal{F}_t)$  would have zero expected P&L. Trend-following is the weakest possible violation of market efficiency — it uses only the price series itself, the single most public dataset in existence — so it is the strategy the efficient-markets hypothesis most obviously forbids.

Three replies, in increasing order of force.

The joint-hypothesis reply. Any test of efficiency is a joint test of efficiency and of an asset-pricing model. If trend-following earns a positive return that is compensation for a risk your model does not contain, efficiency survives. This is logically airtight and empirically almost vacuous — you can always postulate a risk factor — but it does mean “trend-following works” and “markets are efficient” are not formally contradictory.

The Grossman–Stiglitz reply. If prices were perfectly informative, no one would pay to gather information, so prices could not be informative. Equilibrium therefore requires prices to be slightly wrong — enough to pay the marginal information-gatherer’s costs and no more. The equilibrium level of inefficiency is not zero; it is the level at which arbitrage is barely worth doing after costs. Trend-following’s actual measured edge, of order a per-market Sharpe of 0.2–0.4 before costs (\$5.9), is roughly what “barely worth doing” looks like.

The empirical reply. The effect has been documented over very long samples in markets that did not exist when the rules were designed. Hurst, Ooi and Pedersen (2017) report positive trend-following returns across 67 markets back to 1880; Lempérière and co-authors (2014) find a  $t$ -statistic of about 5 since 1960 and about 10 since 1800 on spot commodity and index series, after removing the markets’ own drift. [Contested] Both are practitioner studies by firms that sell the strategy, which is a real discount factor — but they are also the only groups with the data, and their long-horizon results have not been overturned. Set against them, Huang, Li, Wang and Zhou (2020) argue the canonical time-series momentum test is confounded by exactly the drift issue those papers try to remove, and find little evidence of the effect after correcting for it (\$5.3). This disagreement is live and it is about specification, not data.

None of that explains why. Below are the five mechanism families that have been proposed. They are not mutually exclusive, and I think the honest position is that all five contribute, in proportions that vary by asset class and horizon.

### 2.2 Family A — slow information diffusion

[Hypothesis, well-formalised] Information reaches market participants gradually rather than instantaneously. If news arrives to one group at  $t$  and another at  $t + k$ , the price adjusts in steps, and a sequence of partial adjustments in the same direction is precisely positive return autocorrelation.

Hong and Stein (1999) formalise this with two agent types: “newswatchers” who trade on private information but ignore prices, and “momentum traders” who trade on price changes but ignore fundamentals. Slow diffusion among newswatchers produces underreaction, which momentum traders then amplify into overshooting and eventual reversal. The model is attractive because a single assumption generates underreaction at short horizons, momentum at medium ones, and reversal at long ones — the empirically observed sign pattern (§9.2), rather than one piece of it.

What would falsify it: trends should be stronger where information diffuses more slowly — smaller, less-covered, less-liquid markets. This is broadly supported in equities (momentum is stronger in small caps and low-analyst-coverage names) and much weaker as an explanation for trends in Treasury futures or major FX, which are the most information-efficient markets in existence and where trend-following nonetheless works well. That gap is the strongest evidence against diffusion being the whole story.

### 2.3 Family B — behavioural biases

[Hypothesis] Several documented individual-level biases produce sluggish price adjustment in aggregate:

- Anchoring and conservatism. Barberis, Shleifer and Vishny (1998) model investors who update too slowly from a prior, producing underreaction to individual news and overreaction to long streaks.
- Overconfidence and biased self-attribution. Daniel, Hirshleifer and Subrahmanyam (1998) model investors who overweight private signals and attribute confirming public news to their own skill, producing continued overreaction and eventual correction.

- The disposition effect. Investors sell winners too early and hold losers too long. Grinblatt and Han (2005) show this creates a spread between the market price and a reference price, and that the spread predicts returns — the selling pressure on winners retards the adjustment, so the price drifts toward fundamental value slowly rather than jumping.
- Extrapolative expectations. Greenwood and Shleifer (2014) show, from survey data, that investors' expected returns are high after high past returns — the opposite of what a rational model with mean-reverting returns predicts. Barberis and co-authors (2015) build an equilibrium model on this.

The problem with this family is not that the biases are unreal — they are well documented at the individual level — but that the step from individual bias to price impact requires the biased agents to be the marginal price-setters. In Treasury futures they are not. Behavioural explanations do good work in equities and much less in the futures markets where trend-following actually earns most of its money.

## 2.4 Family C — mechanical and flow-driven persistence

[Fact for the mechanism, [Hypothesis] for its share of the effect] This family requires no irrationality at all. Several structural features of markets mechanically generate serially correlated order flow, and serially correlated order flow generates serially correlated returns.

- Metaorder execution. A large institutional order cannot be executed at once; it is split into child orders over hours or days. Each child order pushes price in the same direction. The empirical impact of a metaorder of size  $Q$  scales as  $\sqrt{Q}$  — the “square-root law”, documented across markets and brokers by Tóth and co-authors (2011) — and the impact accumulates over the execution horizon, which is exactly a short-horizon trend.
- Risk-management flows. Volatility targeting, portfolio insurance, stop losses, margin calls, and value-at-risk limits all force selling into declines and buying into rallies. Brunnermeier and Pedersen (2009) formalise the destabilising loop: losses tighten funding, tighter funding forces deleveraging, deleveraging causes losses.
- Systematic rebalancing. Leveraged ETFs must rebalance daily in the direction of the day's move. Target-date and risk-parity funds rebalance on schedules. These are price-taking flows with known signs.
- Central-bank and macro persistence. Policy rates move in cycles lasting quarters to years, and inflation and growth are persistent. Futures markets on rates, bonds and currencies inherit that persistence directly. This is the most plausible single explanation for why trend-following works best in fixed income and FX, and it is not a market-inefficiency story at all.

I regard this family as the most likely primary driver in futures markets, for one reason: it predicts trend-following should work where large slow flows exist regardless of investor sophistication, and that is what the cross-section of results looks like.

## 2.5 Family D — risk transfer and hedging pressure

[Contested] In futures markets there are natural hedgers with inelastic demand — a producer who must sell forward, an airline that must buy fuel. Keynes argued that speculators taking the other side must be compensated, producing a risk premium tied to the direction of hedging pressure. De Roon, Nijman and Veld (2000) find hedging-pressure effects in futures returns across markets.

This explains a return, but it explains a trend only if hedging pressure is itself persistent — which it plausibly is, since a producer's hedging need changes with the slow-moving fundamentals of their business. Under this account trend-following is not exploiting an error; it is being paid to warehouse risk that someone else must shed, and the payment persists as long as the need does.

The attractive feature of this story is that it explains the convexity correctly. A liquidity provider to forced sellers earns a premium and takes tail risk; a trend-follower buys convexity and pays a premium. Those are opposite exposures, which means hedging pressure cannot be the whole explanation — but it may well be a large part of the level of returns in commodities specifically.

## 2.6 Family E — there is no effect, only drift and selection

[Contested] The nihilist position deserves its strongest statement, because it is not obviously wrong.

Most assets have positive expected returns. A trend rule applied to an asset with positive drift will be long most of the time and will therefore earn the drift, and §5.3 shows the standard time-series momentum test is positively biased by exactly this:  $\mathbb{E}[r_{t+1} \cdot \text{sign}(r_{t-L:t})] > 0$  whenever  $\mu > 0$ , even under complete independence. Huang and co-authors (2020) make this argument formally and find the time-series momentum effect largely disappears in their sample once the unconditional mean is controlled for.

Layer on top: the strategy's parameters were selected by decades of practitioner search over the same price histories now used to validate it — the definition of data snooping. Sullivan, Timmermann and White (1999) applied White's Reality Check to a universe of nearly 8,000 technical trading rules and found that rules which looked strongly significant in isolation were not, once the search was accounted for. Brock, Lakonishok and LeBaron (1992) had earlier found technical rules profitable on the Dow 1897–1986; the snooping critique is aimed squarely at results of that kind. Park and Irwin's (2007) survey of the whole technical- analysis literature concludes that early positive results were substantially compromised by data snooping and by ignoring transaction costs.

And CTA returns as actually delivered to investors are far worse than the strategy's paper returns. Bhardwaj, Gorton and Rouwenhorst (2014) find that after fees, investors in commodity trading advisors earned close to nothing, while the funds' gross performance and their persistence were both weak once survivorship and backfill biases were removed.

My read. Families C and D are probably real and probably small; family E is the correct null and is not defeated, only weakened, by the long-history studies; families A and B do genuine work in equities and less elsewhere. The specific claim I would defend is narrow: futures markets exhibit weak positive return autocorrelation at horizons of weeks to months, of a magnitude that supports a diversified gross Sharpe of roughly 0.7–1.0 and a net-of-fee investor experience much worse than that. Anyone claiming more should be asked what their evidence is that survives the drift adjustment.

## 2.7 Telling the families apart

Mechanism talk is cheap unless it makes differential predictions. It does:

| Family                 | Predicts trend is stronger...                              | Predicts trend is weaker...   | Distinguishing test                                                                                 |
|------------------------|------------------------------------------------------------|-------------------------------|-----------------------------------------------------------------------------------------------------|
| A. Slow diffusion      | in less-covered, less-liquid, more-opaque markets          | in the most efficient markets | Cross-sectional: sort markets by information efficiency                                             |
| B. Behavioural         | where retail and unsophisticated flow dominates            | in institutional-only markets | Sort by investor composition; test disposition-effect proxies directly                              |
| C. Mechanical flow     | where large slow flows exist, regardless of sophistication | where flows are small or fast | Use positioning and metaorder data; test whether trend returns concentrate around known flow events |
| D. Hedging pressure    | in physical commodities with concentrated hedgers          | in financial futures          | Commitments-of-Traders positioning as a conditioning variable                                       |
| E. Drift plus snooping | nowhere — the effect is an artefact                        | everywhere, after correction  | Demean returns before testing; out-of-sample on markets and eras not used in rule design            |

The one test that discriminates most cheaply is family E's: demean the returns market-by-market before running any trend test. If your effect survives, you have something; if it does not, you have re-discovered that stocks and bonds went up. This costs one line of code and eliminates the single most common way that trend research fools its author.

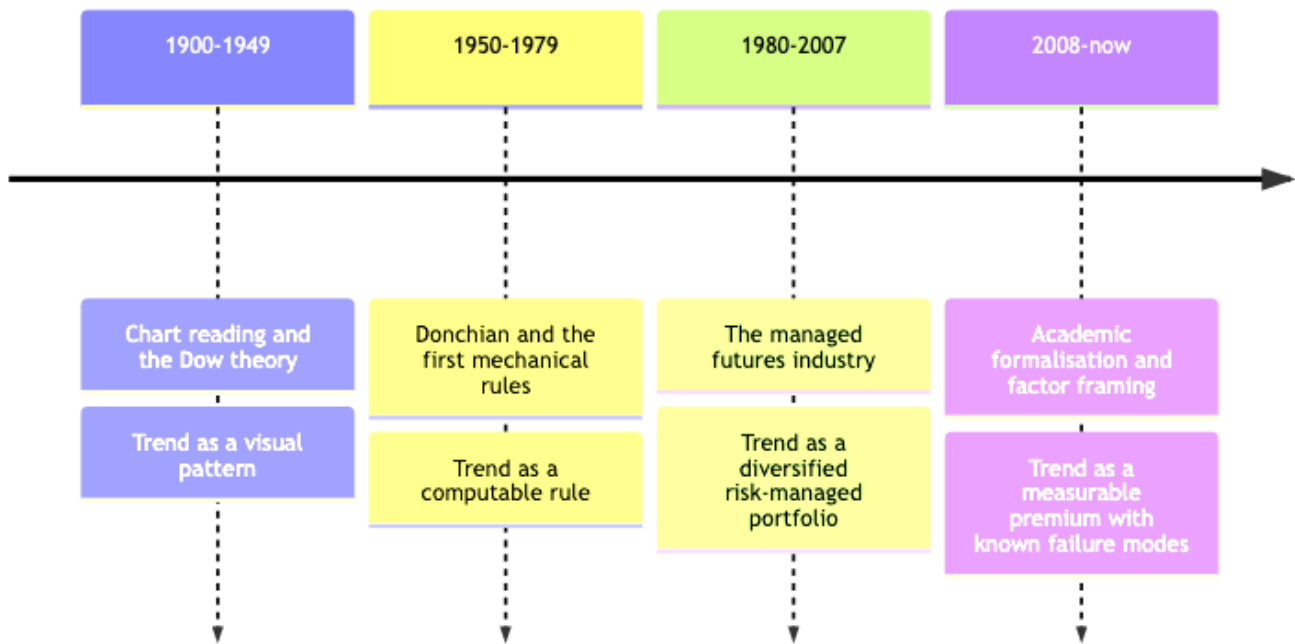
## §2 Key takeaways

1. Trend-following is the weakest possible violation of efficiency — it uses only public prices — so it demands a mechanism, not just a backtest.
2. Grossman–Stiglitz makes a small residual inefficiency a requirement of equilibrium, not an anomaly. The measured size of the trend effect is about what “barely worth arbitraging” should look like.
3. Five mechanism families are on the table: slow diffusion, behavioural bias, mechanical flow, hedging pressure, and “it’s just drift”. They are not mutually exclusive and each explains a different subset of the evidence.
4. Behavioural and diffusion stories do their best work in equities and their worst in Treasury and FX futures — which is where trend-following earns most of its money. Treat that gap as unresolved, not as a detail.
5. Mechanical flow (family C) is my best guess at the primary driver in futures, because it predicts trends where slow flows exist regardless of investor sophistication, which matches the cross-section.
6. The nihilist null — drift plus data snooping — has not been defeated. It has been weakened by long-history evidence produced by interested parties.
7. Demean each market’s returns before testing. It is one line of code and it removes the most common self-deception in this literature.

## 3. How the practice evolved

Trend-following’s history is unusual: the practice ran roughly forty years ahead of the theory, and the two communities barely spoke. Systematic futures traders were running vol-scaled multi-market trend portfolios in the 1970s; the first academic paper to describe that object as a factor appeared in 2012. Reading the history in order is worthwhile mainly because it explains why the standard implementation looks the way it does — most of its features are scar tissue.

# Four eras of trend-following



## 3.1 Era I — Chart reading (1900–1949)

- **Contribution.** The idea that price movements have direction that persists, and that a participant should align with it rather than against it. Dow's editorials, later systematised as "Dow theory", introduced the notion of primary and secondary trends and the confirmation of one index by another.
- **What changed.** Trend became a thing one could have an opinion about systematically, rather than a description applied after the fact.
- **Limitations.** Nothing here was a rule. Every judgement — is this a primary trend, has it been confirmed — required a human, which meant it could not be tested and its results could not be attributed.
- **Lasting influence.** Small, and mostly vocabulary. The specific patterns (head and shoulders, triangles) have not survived quantitative testing; Lo, Mamaysky and Wang (2000) gave them the fairest possible test with kernel smoothing and found weak, mostly non-tradeable, statistical content.

## 3.2 Era II — Mechanical rules (1950–1979)

- **Contribution.** Richard Donchian's moving-average and channel-breakout rules turned trend-following into arithmetic: a rule anyone could compute, on a schedule, without judgement. Donchian's 5- and 20-day moving-average crossover and the " $n$ -day high" breakout are still the reference implementations.
- **What changed.** Once the rule was mechanical, it could be backtested, and its P&L could be attributed to the rule rather than the trader. This is the single most important methodological step in the field's history.
- **Limitations.** Single-market, fixed-size, no volatility scaling, no portfolio construction. Position size was a fixed number of contracts, which means risk varied by a factor of five across markets and across time within one market.
- **Lasting influence.** Enormous. Every rule in §6.1–§6.5 is a Donchian descendant, and the breakout rule in particular is essentially unchanged.

## 3.3 Era III — The industry (1980–2007)

- **Contribution.** The construction of trend-following as a portfolio discipline: many markets, positions sized by volatility, risk budgeted at the book level, systematic roll and execution. The firms — Campbell, Millburn, AHL, Chesapeake, Winton, Aspect, and the Dennis–Eckhardt "Turtle" cohort — developed this largely in private and largely without publishing.

- What changed. The realisation that the edge per market is small and the portfolio effect is most of the return. A per-market Sharpe of 0.3 across 50 weakly correlated markets is a portfolio Sharpe near 1.0 (§5.9); no single market's rule needed to be good.
- Limitations. Almost none of it was published, so the evidence base was track records — which carry survivorship, selection, and fee opacity. Bhardwaj, Gorton and Rouwenhorst (2014) later showed how much those biases were hiding.
- Lasting influence. This is where the modern implementation was invented. Volatility targeting, correlation-aware sizing, multi-timescale ensembles, and systematic roll all date from this era, and §8 is essentially a description of what these firms worked out.

### 3.4 Era IV — Formalisation (2008–present)

- Contribution. Academic and quantitative-practitioner work that made the object measurable. Moskowitz, Ooi and Pedersen (2012) introduced “time-series momentum” as a factor across 58 futures markets. Fung and Hsieh (2001) had already shown that trend-follower returns load on lookback straddles rather than on linear asset exposures. Potters and Bouchaud (2006), Bruder and Gausssel (2011) and Dao and co-authors (2016) derived the P&L decomposition that §5 reconstructs. Levine and Pedersen (2016) and Beekhuizen and Hallerbach (2017) showed that the various trend rules are one filter with different weights.
- What changed. Trend-following stopped being a strategy with a track record and became a factor with a functional form, a risk decomposition, a set of known failure modes, and a price. That is also when its fees collapsed.
- Limitations. The formalisation arrived alongside a decade of poor realised performance (2009–2019), and it is genuinely unclear how much of the formalisation is description of a real premium and how much is a well-specified account of a data-snooped one (§9.4).
- Lasting influence. Ongoing. The current frontier — changepoint-aware signals, learned response functions, cost-aware dynamic portfolios — is all built on this era's framing.

#### §3 Key takeaways

1. The practice preceded the theory by roughly forty years; almost everything in a modern system was invented by practitioners before it was described.
2. Donchian's contribution was not a better indicator but mechanisation — which is what made backtesting and attribution possible at all.
3. The industry's key insight was that the per-market edge is small and the portfolio effect supplies most of the Sharpe. Design for the portfolio.
4. Almost all Era III evidence was track records, which is why the survivorship and fee critiques of the 2010s landed as hard as they did.
5. The 2008-onward formalisation coincided with a decade of weak performance. Whether that is coincidence or the effect of the formalisation is the open question of §9.

## 4. Foundational references

Annotated, grouped by how they should be read. Every entry links to a freely readable copy where one exists.

### 4.1 The P&L and payoff structure

- Fung, W. & Hsieh, D. A. (2001). “The Risk in Hedge Fund Strategies: Theory and Evidence from Trend Followers.” *Review of Financial Studies* 14(2), 313–341. [DOI] — The paper that established trend-follower returns as option-like, and modelled them with lookback straddles. The origin of every “crisis alpha” claim.
- Potters, M. & Bouchaud, J.-P. (2006). “Trend followers lose more often than they gain.” *Wilmott Magazine*. — Short, exact, and the best cure for hit-rate confusion. Average gain per trade is zero on a random walk while the win rate falls below one half.
- Bruder, B. & Gausssel, N. (2011). “Risk-Return Analysis of Dynamic Investment Strategies.” *Lyxor White Paper Series* 7. — The continuous-time P&L decomposition into a convex “option profile” and a running cost. §5.5 derives the discrete analogue.

- Dao, T.-L., Nguyen, T.-T., Deremble, C., Lempérière, Y., Bouchaud, J.-P. & Potters, M. (2016). “Tail protection for long investors: Trend convexity at work.” *Journal of Investment Strategies*. — Shows trend P&L equals the difference between long- and short-horizon realised variance. This is the cleanest statement of the spine of this document.

## 4.2 The empirical case

- Moskowitz, T. J., Ooi, Y. H. & Pedersen, L. H. (2012). “Time Series Momentum.” *Journal of Financial Economics* 104(2), 228–250. — The paper that made trend-following academically respectable. 58 futures, 1965–2009, one to twelve month horizons, with partial reversal beyond.
- Hurst, B., Ooi, Y. H. & Pedersen, L. H. (2017). “A Century of Evidence on Trend-Following Investing.” *Journal of Portfolio Management* 44(1), 15–29. — 1880–2016, 67 markets. [Contested] — AQR sells this strategy; read it alongside §2.6.
- Lempérière, Y., Deremble, C., Seager, P., Potters, M. & Bouchaud, J.-P. (2014). “Two centuries of trend following.” *Journal of Investment Strategies*. — Independent long-history evidence from a different interested party (CFM), which matters: two firms with different books reaching the same conclusion is worth more than either alone.
- Babu, A., Levine, A., Ooi, Y. H., Pedersen, L. H. & Stamelos, E. (2020). “Trends Everywhere.” *Journal of Investment Management* 18(1), 52–68. — Extends the evidence to less-traded markets and alternative instruments.
- Hamill, C., Rattray, S. & Van Hemert, O. (2016). “Trend Following: Equity and Bond Crisis Alpha.” Man AHL working paper. — The crisis-alpha claim, tested over a long sample and separated by asset class. [Contested] — interested party again.

## 4.3 The critiques — read these next to the section above

- Huang, D., Li, J., Wang, L. & Zhou, G. (2020). “Time series momentum: Is it there?” *Journal of Financial Economics* 135(3), 774–794. — The drift-confound critique. Directly determines how you must specify your own tests (§5.3, §10.2).
- Bhardwaj, G., Gorton, G. B. & Rouwenhorst, K. G. (2014). “Fooing Some of the People All of the Time: The Inefficient Performance and Persistence of Commodity Trading Advisors.” *Review of Financial Studies* 27(11), 3099–3132. — What investors actually received, as opposed to what the strategy produced.
- Sullivan, R., Timmermann, A. & White, H. (1999). “Data-Snooping, Technical Trading Rule Performance, and the Bootstrap.” *Journal of Finance* 54(5), 1647–1691. — The reality-check methodology, applied to a universe of thousands of technical rules.
- Park, C.-H. & Irwin, S. H. (2007). “What Do We Know About the Profitability of Technical Analysis?” *Journal of Economic Surveys* 21(4), 786–826. — The comprehensive survey. Sober and useful precisely because it is not written by anyone with a book.
- Kim, A. Y., Tse, Y. & Wald, J. K. (2016). “Time series momentum and volatility scaling.” *Journal of Financial Markets* 30, 103–124. — Argues that much of the reported time-series momentum premium comes from the volatility scaling rather than from the trend signal. Uncomfortable, and important (§8.3).

## 4.4 Implementation and portfolio construction

- Levine, A. & Pedersen, L. H. (2016). “Which Trend Is Your Friend?” *Financial Analysts Journal* 72(3), 51–66. — Relates time-series momentum and moving-average rules through their common weighting of past returns.
- Beekhuizen, P. & Hallerbach, W. G. (2017). “Uncovering Trend Rules.” *Journal of Alternative Investments* 20(2), 28–38. — Derives the return-space weights implied by moving-average rules and shows some popular rules have inverted decay or hidden mean-reversion. The direct antecedent of §7.2.
- Zakamulin, V. (2017). *Market Timing with Moving Averages: The Anatomy and Performance of Trading Rules*. Palgrave Macmillan. [paywalled] — Book-length treatment of the same decomposition, plus a large and unusually honest empirical study.
- Baltas, N. & Kosowski, R. (2013). “Demystifying Time-Series Momentum Strategies: Volatility Estimators, Trading Rules and Pairwise Correlations.” *Journal of Derivatives & Hedge Funds* 19(4), 289–310. — The most directly useful implementation paper in the list: volatility estimator choice, turnover reduction, and a correlation-adjusted leverage mechanism.
- Gârleanu, N. & Pedersen, L. H. (2013). “Dynamic Trading with Predictable Returns and Transaction Costs.” *Journal of Finance* 68(6), 2309–2340. — The closed-form optimal policy with costs: aim in front of the target, trade partially toward it. Section 8.7 is an application.
- Harvey, C. R., Hoyle, E., Korgaonkar, R., Rattray, S., Sargaison, M. & Van Hemert, O. (2018). “The Impact of Volatility Targeting.” *Journal of Portfolio Management* 45(1), 14–33. — What volatility targeting does to Sharpe, drawdown and skew, by asset class.

- Carver, R. (2015). *Systematic Trading*. Harriman House. — The most practically complete public description of a full trend system, from signal scaling through to position rounding. Opinionated and specific.

## 4.5 Foundational statistical machinery

- Muth, J. F. (1960). “Optimal Properties of Exponentially Weighted Forecasts.” *Journal of the American Statistical Association* 55(290), 299–306. [paywalled] — The EWMA is the minimum-mean-squared-error forecast for a random walk observed in noise. This is why the exponential kernel is everywhere (§5.8).
- Lo, A. W. & MacKinlay, A. C. (1988). “Stock Market Prices Do Not Follow Random Walks: Evidence from a Simple Specification Test.” *Review of Financial Studies* 1(1), 41–66. — The variance-ratio test, which §5.4 shows is the same object as a trend rule’s expected P&L.
- Kim, S.-J., Koh, K., Boyd, S. & Gorinevsky, D. (2009). “ $\ell_1$  Trend Filtering.” *SIAM Review* 51(2), 339–360. — Piecewise-linear trend extraction as a convex program. The principled version of “the trend changed”.
- Kaminski, K. M. & Lo, A. W. (2014). “When Do Stop-Loss Rules Stop Losses?” *Journal of Financial Markets* 18, 234–254. — Stops help only when returns are positively autocorrelated; under a random walk they cost. §11.3.
- Bailey, D. H. & López de Prado, M. (2014). “The Deflated Sharpe Ratio.” *Journal of Portfolio Management* 40(5), 94–107. — How to discount a Sharpe ratio for the number of trials that produced it.

## 4.6 If you only read six things

1. Moskowitz, Ooi & Pedersen (2012) — the empirical object, stated cleanly.
2. Huang, Li, Wang & Zhou (2020) — immediately after, so you never run an unadjusted test.
3. Dao et al. (2016) — what the P&L actually is.
4. Potters & Bouchaud (2006) — six pages that fix your intuition about the payoff distribution permanently.
5. Beekhuizen & Hallerbach (2017) — after which you will stop believing that different moving-average rules are different strategies.
6. Carver (2015) — because everything above still leaves you a long way from a working system, and this closes most of the gap.

# 5. The mathematics of trend-following P&L

This section answers “what does a trend-follower actually earn?” — exactly, not approximately. Four results, each derived rather than asserted:

- §5.2: expected P&L is a weighted sum of return autocovariances.
- §5.4: for the canonical rule, that sum is exactly half the variance-ratio excess. Trend-following and the variance-ratio test are the same statistic.
- §5.5: an exact algebraic identity splitting realised P&L into a convex term, a “trend energy” term, and a realised-variance cost. No distributional assumptions at all.
- §5.8: the optimal kernel, and the surprising fact that at realistic signal-to-noise ratios the lookback is set almost entirely by how long trends last, not by how strong they are.

## 5.1 Setup

Fix one market. Positions are set from information through  $t$  and held for one bar, so the P&L over  $T$  bars is

$$\Pi_T = \sum_{t=0}^{T-1} \pi_t r_{t+1}, \quad \pi_t \in \mathcal{F}_t.$$

Two simplifications, both harmless here and both revisited later. First,  $\pi_t$  is an exposure in units of notional, so  $\pi_t r_{t+1}$  is a return on capital times that exposure; costs are absent and enter in §8.8. Second, I take the signal to be a linear filter of past returns,

$$s_t = \sum_{k \geq 0} w_k r_{t-k},$$

which §7.2 shows covers the moving-average family exactly, and which the breakout and change-point rules of §6.5 and §6.8 violate.

## 5.2 Expected P&L is a weighted sum of autocovariances

Take  $\pi_t = s_t$  — the linear response, no risk scaling — and assume returns are covariance-stationary with mean  $\mu$  and autocovariances  $\gamma_k = \text{Cov}(r_t, r_{t+k})$ . Then the expected P&L per bar is

$$\mathbb{E}[\pi_t r_{t+1}] = \sum_{k \geq 0} w_k \mathbb{E}[r_{t-k} r_{t+1}] = \sum_{k \geq 0} w_k (\gamma_{k+1} + \mu^2) = \underbrace{\sum_{k \geq 0} w_k \gamma_{k+1}}_{\text{predictability}} + \underbrace{\mu^2 \sum_{k \geq 0} w_k}_{\text{drift}}.$$

The second step uses  $\mathbb{E}[XY] = \text{Cov}(X, Y) + \mathbb{E}X \mathbb{E}Y$  with  $\mathbb{E}[r_t] = \mu$  for every  $t$ ; the lag between  $r_{t-k}$  and  $r_{t+1}$  is  $k+1$ , which is why the autocovariance index is shifted.

This little identity is the whole story of §5, and it says three things.

First, trend-following is long autocovariance. The strategy's edge is a  $w$ -weighted inner product with the autocovariance function. If  $\gamma_k = 0$  for all  $k \geq 1$  — a martingale difference sequence — the predictability term is exactly zero regardless of how clever the kernel is. There is no filter design that extracts edge from an unpredictable series, and any backtest suggesting otherwise is measuring drift, cost errors, or luck.

Second, the kernel must be matched to where the predictability lives. The inner product  $\sum_k w_k \gamma_{k+1}$  is large only when  $w$  has mass at the lags where  $\gamma$  does. A slow kernel run on a series whose autocorrelation decays within a day earns nothing — not because the series is unpredictable, but because the filter is looking in the wrong place. I saw this directly in simulation: an EWMA(32, 128) rule applied to an AR(1) return process with  $\phi = 0.03$  earned a Sharpe of 0.00, because that kernel's weight is concentrated around lag 30 where  $\phi^{30} \approx 10^{-47}$ . Matching the kernel's centre of mass to the autocorrelation's decay length is the single design decision that cannot be fudged.

Third, the drift term is a trap. Since a trend kernel has  $\sum_k w_k > 0$  by construction, the drift term  $\mu^2 \sum_k w_k$  is strictly positive whenever  $\mu \neq 0$  — of either sign. That is the subject of the next subsection.

## 5.3 The drift confound, stated precisely

Consider the canonical time-series momentum test: regress  $r_{t+1}$  on  $\text{sign}(r_{t-L:t})$ , where  $r_{t-L:t} = \sum_{i=1}^L r_{t-L+i}$  is the cumulative return over the lookback. The estimand is

$$\mathbb{E}[r_{t+1} \cdot \text{sign}(r_{t-L:t})].$$

Suppose returns are completely independent — no predictability of any kind — but with mean  $\mu > 0$ . Write  $R = r_{t-L:t}$ , which is independent of  $r_{t+1}$ . Then

$$\mathbb{E}[r_{t+1} \text{sign}(R)] = \mathbb{E}[r_{t+1}] \cdot \mathbb{E}[\text{sign}(R)] = \mu(2\mathbb{P}(R > 0) - 1) > 0$$

because  $\mu > 0$  implies  $\mathbb{P}(R > 0) > 1/2$ . The test statistic is positive under the null. For a daily market with  $\mu/\sigma = 0.03$  per bar and  $L = 252$ ,  $\mathbb{P}(R > 0) = \Phi(0.03\sqrt{252}) = \Phi(0.476) \approx 0.68$ , so the bias is  $0.36\mu$  — a substantial fraction of the drift itself, appearing as if it were predictability.

This is the Huang, Li, Wang and Zhou (2020) critique. [Contested] Their conclusion — that little time-series momentum survives the correction — is disputed by Moskowitz, Ooi and Pedersen and by subsequent replications, and the disagreement is about how to demean (in-sample full-period mean? expanding window? cross-sectionally?) rather than about the algebra, which is not in question.

What to do about it. Three options, in increasing order of conservatism:

1. Demean each market's returns using an expanding-window estimate before computing signals and P&L. Cheap, and removes most of the bias without look-ahead.
2. Include the passive long as a benchmark. Report trend P&L as an alpha against a constant-long position in the same market at the same average exposure. If the alpha is zero you have re-derived buy-and-hold.

3. Test on demeaned bootstrap resamples. Under a block bootstrap that preserves the marginal distribution but destroys serial dependence, a genuine trend effect should vanish; if it does not, your test statistic is picking up something other than serial dependence (§10.4).

I would do all three, and I would treat any trend result reported without at least the first as uninformative.

## 5.4 The exact variance-ratio identity

Now the result that unifies the practitioner and academic literatures.

The variance ratio at horizon  $q$  compares the variance of a  $q$ -bar return to  $q$  times the variance of a one-bar return:

$$\text{VR}(q) = \frac{\text{Var}(\sum_{i=1}^q r_i)}{q \gamma_0} = 1 + 2 \sum_{k=1}^{q-1} \left(1 - \frac{k}{q}\right) \rho_k.$$

Under a random walk  $\text{VR}(q) = 1$  for all  $q$ ; trending series give  $\text{VR} > 1$ , mean-reverting series  $\text{VR} < 1$ . It is the standard non-parametric test for departures from the random walk (Lo and MacKinlay, 1988).

Now take the most-used practitioner rule in existence — price minus its  $n$ -bar moving average — and compute its expected P&L. Write  $\text{MA}_n(p)_t = \frac{1}{n} \sum_{i=0}^{n-1} p_{t-i}$ . Then

$$p_t - \text{MA}_n(p)_t = \frac{1}{n} \sum_{i=0}^{n-1} (p_t - p_{t-i}) = \frac{1}{n} \sum_{i=0}^{n-1} \sum_{j=0}^{i-1} r_{t-j} = \sum_{j \geq 0} \frac{(n-1-j)^+}{n} r_{t-j},$$

using  $(z)^+ = \max(z, 0)$ ; the last step counts, for each lag  $j$ , how many of the  $n$  inner sums contain  $r_{t-j}$ . So the rule is a linear filter with a triangular kernel  $w_j = (n-1-j)^+/n$ . Substituting into §5.2 with  $\mu = 0$ :

$$\mathbb{E} \left[ (p_t - \text{MA}_n(p)_t) r_{t+1} \right] = \sum_{j=0}^{n-2} \frac{n-1-j}{n} \gamma_{j+1} = \gamma_0 \sum_{k=1}^{n-1} \left(1 - \frac{k}{n}\right) \rho_k = \boxed{\frac{\gamma_0}{2} (\text{VR}(n) - 1)}$$

where the last equality is just the variance-ratio definition rearranged.

This is exact, not an approximation. The expected per-bar P&L of the price-versus-moving-average rule with window  $n$  equals one half the return variance times the variance-ratio excess at horizon  $n$ . I verified it numerically on simulated AR(1) paths: empirical  $1.630 \times 10^{-5}$  against a theoretical  $1.630 \times 10^{-5}$ .

Three consequences worth sitting with:

1. The trend-follower's P&L and the econometrician's random-walk test are the same object. A firm running a 200-day moving-average rule is, in expectation, collecting  $\frac{1}{2} \sigma^2 (\text{VR}(200) - 1)$  per day. If you want to know whether trend-following should work in a market, estimate its variance-ratio profile; you do not need to backtest anything.
2. The horizon is not a free parameter, it is the horizon of the statistic you are betting on. Choosing  $n$  is choosing which  $\text{VR}(q)$  you are long. Since VR is typically above one at some horizons and below one at others (§9.2), the sign of your expected P&L is a function of  $n$  in a way that has nothing to do with the quality of your implementation.
3. It explains why trend-followers and volatility traders are natural counterparties. Being long  $\text{VR}(n) - 1$  is being long long-horizon variance and short short-horizon variance. Someone selling a variance term-structure spread is on the other side.

## 5.5 The exact P&L decomposition: long trend energy, short realised variance

The previous result is about expectations. This one is about the realised path, and it holds identically — for any return sequence whatsoever, with no assumptions about stationarity, distribution, or independence.

Let  $x_t$  be the EWMA trend estimate,  $x_{t+1} = (1 - \alpha)x_t + \alpha r_{t+1}$ , and take the linear-response position  $\pi_t = x_t$ . Square the recursion:

$$x_{t+1}^2 = (1 - \alpha)^2 x_t^2 + 2\alpha(1 - \alpha) x_t r_{t+1} + \alpha^2 r_{t+1}^2.$$

Solve for the P&L increment  $x_t r_{t+1}$  and sum over  $t = 0, \dots, T-1$ . The  $x^2$  terms telescope — writing  $x_{t+1}^2 - (1-\alpha)^2 x_t^2 = (x_{t+1}^2 - x_t^2) + \alpha(2-\alpha)x_t^2$  and summing the first part gives  $x_T^2 - x_0^2$  — which leaves the identity

$$\sum_{t=0}^{T-1} x_t r_{t+1} = \underbrace{\frac{x_T^2 - x_0^2}{2\alpha(1-\alpha)}}_{\text{(A) convexity}} + \underbrace{\frac{2-\alpha}{2(1-\alpha)} \sum_{t=0}^{T-1} x_t^2}_{\text{(B) trend energy}} - \underbrace{\frac{\alpha}{2(1-\alpha)} \sum_{t=1}^T r_t^2}_{\text{(C) realised variance}}$$

I verified this numerically: over 5,000 bars the left side was  $1.1880423066819878 \times 10^{-3}$  and the right side  $1.1880423066819926 \times 10^{-3}$  — equal to machine precision, as an algebraic identity must be.

Read the three terms.

(A) Convexity. A non-negative function of where the trend estimate ends up, minus where it started. This is the option payoff: you profit if the trend estimate has moved away from zero, in either direction. It does not grow with  $T$  — it is a boundary term — so it contributes nothing to the long-run rate of return, but it is exactly the term that pays out in a crisis, when  $|x_T|$  is large. This is the algebraic content of “crisis alpha”.

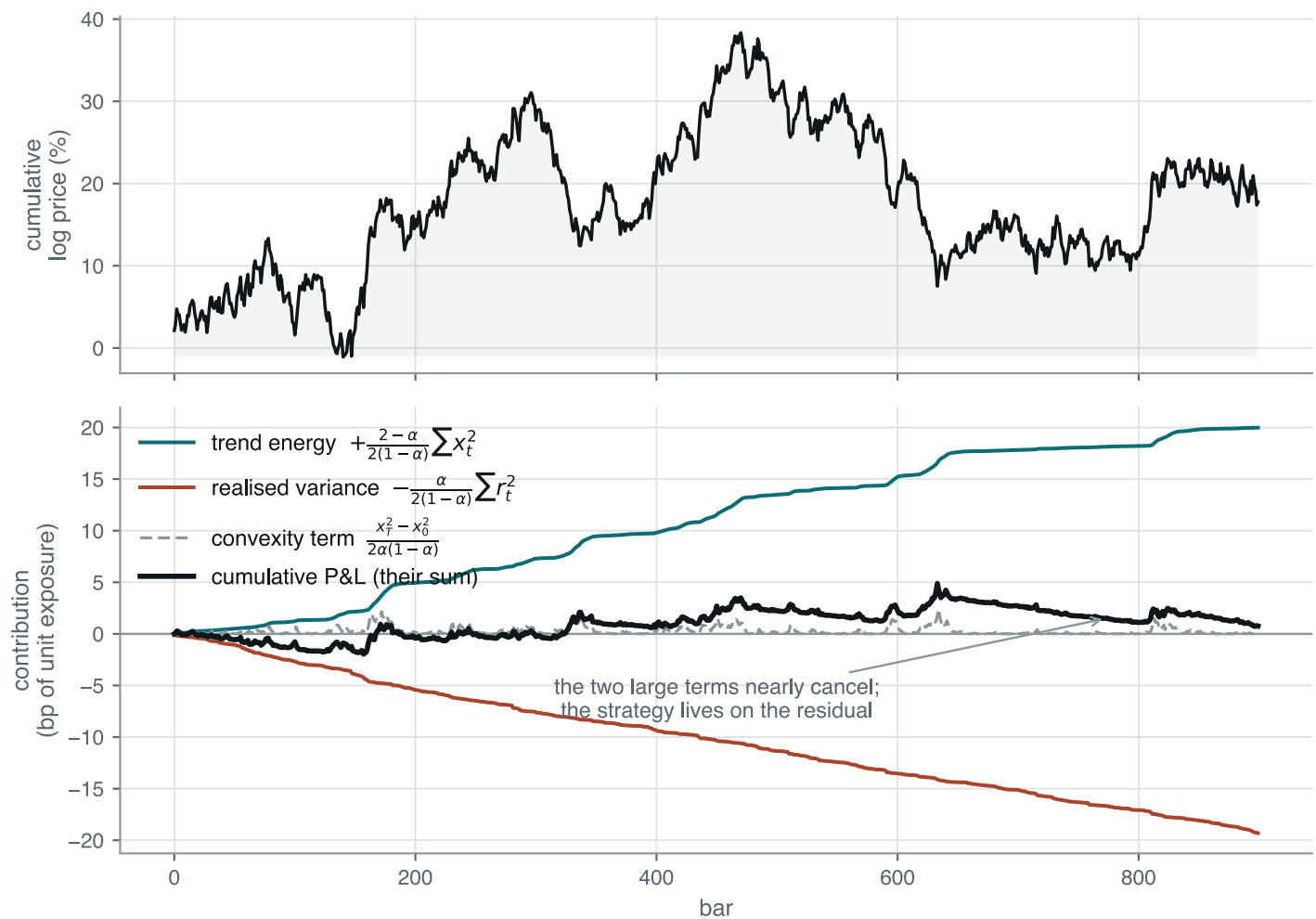
(B) Trend energy. The accumulated squared trend estimate. Always positive, grows linearly in  $T$ . This is what you collect while a trend persists: the larger and the more sustained the filtered trend, the more this pays.

(C) Realised variance. The accumulated squared returns, entering with a minus sign, scaled by  $\alpha$ . This is the premium. A faster filter (larger  $\alpha$ ) pays more of it. It is the exact, quantified version of “whipsaw”.

So: a trend-follower is long the energy of the filtered trend and short the realised variance of the underlying, in a ratio fixed by the filter speed.

That framing immediately explains the strategy’s behaviour. Under a random walk the two large terms cancel in expectation — with iid returns of variance  $\sigma^2$ , steady-state  $\mathbb{E}[x_t^2] = \alpha\sigma^2/(2-\alpha)$ , so term (B) contributes  $\alpha\sigma^2/(2(1-\alpha))$  per bar and term (C) subtracts exactly the same, leaving zero, as it must. Under a trending process,  $x_t^2$  is inflated by the drift while  $r_t^2$  is not, and the residual is the edge.

One path: hidden drift plus noise, and where the trend rule's money comes from



The figure is the identity drawn on one simulated path. Note the scale: the two large terms reach roughly +20 and -19 basis points of cumulative contribution, and the strategy's entire P&L is the  $\sim 1$  basis point residual between them. Trend-following is a small difference between two large numbers, which is precisely why it is so sensitive to costs, to the volatility estimate, and to the filter speed — all three move one of the two large terms.

Dao and co-authors (2016) reach the equivalent conclusion in continuous time and state it memorably: trend P&L is the difference between long-horizon and short-horizon realised variance. The discrete identity above is the same statement with  $\alpha$  setting which horizons.

## 5.6 The option analogy, made precise

Fung and Hsieh (2001) modelled trend-follower returns as a portfolio of lookback straddles — options paying the maximum price move over a period — and found this explained CTA returns far better than linear asset exposures. The analogy is genuinely useful, and it is important to know where it is exact and where it is not.

| Property                               | Long straddle            | Trend rule                                               | Same?                         |
|----------------------------------------|--------------------------|----------------------------------------------------------|-------------------------------|
| Payoff convex in the underlying's move | Yes                      | Yes (§1.5 figure)                                        | Exact                         |
| Premium paid                           | Once, up front, known    | Continuously, as $-\frac{\alpha}{2(1-\alpha)}\sum r_t^2$ | Analogous, not identical      |
| Long volatility                        | Long implied vol         | Long trend vol, short realised vol                       | Opposite sign on realised vol |
| Payoff depends on path                 | Only at expiry (vanilla) | Entirely path-dependent                                  | Different                     |

| Property     | Long straddle                  | Trend rule                                            | Same?                                    |
|--------------|--------------------------------|-------------------------------------------------------|------------------------------------------|
| Maximum loss | Bounded by premium             | Unbounded in principle; bounded in practice by sizing | Different                                |
| Gap risk     | Protected — you own the option | Exposed — you must trade to adjust                    | Different, and this is the important one |

The row that matters is the last. A straddle holder is protected against a discontinuous move because the option's payoff is already contracted. A trend-follower's convexity is synthetic: it is manufactured by trading, and manufacturing it requires the market to be open and continuous. A gap through your position — a limit-locked commodity, a currency de-peg, a weekend — is exactly the state in which the replication fails. The trend-follower is short gap risk and long diffusive convexity, which is a very different object from a straddle even though the smooth-path payoffs match. §11.5 returns to this.

A second caveat, less often stated: the convexity in the §1.5 figure is convexity with respect to the underlying's realised move, not with respect to the equity market. Trend-following is long convexity in each of its markets separately. It delivers "crisis alpha" only when a crisis produces a sustained directional move that its filters can catch — the 2008 equity decline, which took months, rather than the February 2018 volatility spike, which took days.

## 5.7 The trade-level distribution

Combining §5.5's convexity with §1.5's simulation gives the characteristic shape of trend-following returns, which is worth stating as calibration.

From my own simulation of a slow trend rule on a weakly autocorrelated process, sampled over 400,000 bars and segmented into trades at each position flip:

| Statistic                             | Value                                    | Interpretation                             |
|---------------------------------------|------------------------------------------|--------------------------------------------|
| Hit rate                              | 25%                                      | Three quarters of trades lose              |
| Mean win / mean loss                  | 3.3                                      | Winners are three times the size of losers |
| Trade P&L skew                        | +3.6                                     | Heavily right-skewed                       |
| Contribution of the best 5% of trades | $8.2\times$ the strategy's total net P&L | The rest, in aggregate, lose               |

The last row is the one to internalise. The best 5% of trades earned more than eight times the strategy's entire net profit; everything else lost the difference. That is not a pathology, it is the design: the strategy takes many cheap small losses to be present for the rare large moves.

Three practical consequences:

1. You cannot evaluate a trend system on a small number of trades. With this distribution, the sample mean converges very slowly. The effective sample size is closer to the number of large moves in the period than to the number of trades.
2. Any modification that truncates the right tail is likely to destroy the strategy — profit targets being the canonical example. Symmetrically, modifications that truncate the left tail (stops) are much less harmful, which is the asymmetry §11.3 is about.
3. The reported hit rate is a design choice. As §1.5 showed, moving from a linear to a sign response takes the win rate from 41% to 50% with no change in expected return. A manager advertising a high hit rate is telling you about their response function.

## 5.8 What sets the lookback: Muth, Kalman, and the persistence timescale

So far the kernel has been arbitrary. What is the right one?

Assume returns are a slowly varying hidden drift plus noise — the local level model, which is the minimal generative model in which trend-following is the correct strategy:

$$r_t = \mu_t + \varepsilon_t, \quad \mu_t = \phi \mu_{t-1} + \eta_t, \quad \varepsilon_t \sim (0, \sigma_\varepsilon^2), \quad \eta_t \sim (0, \sigma_\eta^2)$$

with  $\phi$  close to 1, so  $\tau = 1/(1 - \phi)$  is the trend's persistence in bars. The two parameters that matter are  $\tau$  and the signal-to-noise ratio  $\theta = \text{sd}(\mu_t)/\sigma_\varepsilon$ .

The optimal estimator of  $\mu_t$  given  $\mathcal{F}_t$  is the Kalman filter, and for this one-dimensional model it has a closed form. Its steady-state gain  $K$  solves the algebraic Riccati equation, and the filter recursion is

$$\hat{\mu}_t = \phi(1 - K) \hat{\mu}_{t-1} + K r_t,$$

which is an EWMA with decay  $\alpha^* = 1 - \phi(1 - K)$ . When  $\phi = 1$  (a random-walk trend) this reduces to  $\alpha^* = K$  and Muth's (1960) result: the exponentially weighted moving average is the minimum-mean-squared-error forecast for a random walk observed in noise. That is why the exponential kernel is ubiquitous — not convention, but optimality under the simplest model in which trends exist at all.

Expanding  $\alpha^*$  for small  $K$  gives the practically important form:

$$\alpha^* = 1 - \phi(1 - K) \approx \underbrace{\frac{1}{\tau}}_{\text{trend decay}} + \underbrace{K}_{\text{signal strength}}, \quad \text{span} = \frac{2}{\alpha^*} - 1.$$

At realistic signal-to-noise ratios  $K$  is tiny — of order  $10^{-3}$  — so  $\alpha^* \approx 1/\tau$  and the second term is nearly irrelevant.

The optimal lookback is set by how long trends last, not by how strong they are.

This is a genuinely useful result, and it has a clear operational reading: do not tune the lookback by maximising backtest P&L, which is a noisy objective with thousands of effective trials. Estimate the persistence timescale of the process — from the autocorrelation decay, the variance-ratio profile, or the economic mechanism you believe in — and set the span to roughly  $2\tau$ .

I checked the formula against brute-force optimisation over EWMA spans, on simulated paths of the model above:

| $\tau$ (bars) | $\theta$ | $\alpha^*$ from Kalman | Implied span | Best span by search | Sharpe at optimum |
|---------------|----------|------------------------|--------------|---------------------|-------------------|
| 60            | 0.05     | 0.0190                 | 104          | 101                 | 0.20              |
| 120           | 0.05     | 0.0105                 | 189          | 181                 | 0.28              |
| 250           | 0.05     | 0.0060                 | 333          | 298                 | 0.36              |
| 250           | 0.10     | 0.0098                 | 204          | 214                 | 1.01              |
| 500           | 0.07     | 0.0048                 | 411          | 415                 | 0.74              |

The agreement is close across the range. Note the third and fourth rows: holding  $\tau$  fixed at 250 and doubling the signal-to-noise ratio changes the optimal span from 333 to 204 — a real effect, but small compared to what changing  $\tau$  does.

The optimum is flat, and that is the practically important part

Optimal is one thing; sensitivity is another. On the  $\tau = 250, \theta = 0.05$  process, sweeping the EWMA span gives:

|           |       |       |       |       |       |       |       |       |
|-----------|-------|-------|-------|-------|-------|-------|-------|-------|
| Span      | 5     | 10    | 20    | 40    | 80    | 160   | 320   | 640   |
| Sharpe    | 0.085 | 0.120 | 0.167 | 0.221 | 0.280 | 0.331 | 0.352 | 0.334 |
| % of best | 24    | 34    | 47    | 63    | 79    | 94    | 100   | 95    |

Being wrong by a factor of two costs about 5% of the Sharpe. Being wrong by a factor of sixteen costs half of it. The lookback is therefore worth getting approximately right and not worth optimising — which is fortunate, because the data cannot support optimising it (§10.5).

This also settles a common practitioner claim. [Contested] Multi-timescale ensembles — averaging signals across several spans — are usually justified as adding return. In my simulations they do not: an equal-weighted ensemble of six spans from 20 to 640 achieved 93% of the best single span on a stationary process and 97% on a process whose  $\tau$  switched among 40, 150 and 600. The ensemble is never better than the best span. What it does is convert “choose right or lose half your Sharpe” into “always get about 95%”. That is insurance against mis-specification, which is worth buying — but it should be sold as insurance, not as alpha.

## 5.9 Calibration: what edge corresponds to what Sharpe

Putting numbers on the model makes the whole field legible. Take the local level model with  $\tau = 250$  days and  $\theta = 0.05$ , filtered with its optimal EWMA. The resulting process has:

| Quantity                                               | Value           |
|--------------------------------------------------------|-----------------|
| Lag-1 return autocorrelation $\rho_1$                  | $\approx 0.005$ |
| VR(60)                                                 | $\approx 1.13$  |
| Per-market annualised Sharpe of the optimal filter     | $\approx 0.36$  |
| $t$ -statistic on $\rho_1$ from 10 years of daily data | $\approx 0.25$  |
| Portfolio Sharpe, 50 markets at $\bar{\rho} = 0.15$    | $\approx 0.88$  |

The portfolio step uses the standard aggregation identity:  $N$  strategies each with Sharpe  $S$  and average pairwise correlation  $\bar{\rho}$  combine to

$$S_{\text{portfolio}} = S \sqrt{\frac{N}{1 + (N - 1)\bar{\rho}}}.$$

For  $N = 50$ ,  $\bar{\rho} = 0.15$  the multiplier is 2.45.

Read the table downward and the field snaps into focus.

[Fact] The predictability underlying a diversified trend program with a long-run Sharpe near 1.0 is a daily return autocorrelation of about half a percent — which, in ten years of one market’s data, produces a  $t$ -statistic of 0.25. The effect that supports an entire industry is individually undetectable. Three things follow:

1. You cannot validate a trend signal market by market. Anyone who tells you their rule “works on gold” over ten years is reporting noise, whichever sign they report.
2. Diversification is not a refinement, it is the mechanism. The  $\sqrt{N/(1 + (N - 1)\bar{\rho})}$  multiplier is where the Sharpe comes from; the per-market signal only has to be marginally positive on average.
3. The academic literature’s difficulty rejecting the random walk and the industry’s ability to make money are not in tension. They are the same fact viewed at different levels of aggregation.

The correlation term matters more than the count. Going from 50 markets to 100 at  $\bar{\rho} = 0.15$  improves the multiplier from 2.45 to 2.51 — nothing. Reducing  $\bar{\rho}$  from 0.15 to 0.05 at  $N = 50$  improves it from 2.45 to 3.81 — a 55% increase in Sharpe. Finding genuinely uncorrelated markets is worth far more than finding more markets, which is §8.6’s whole subject.

Finally, the sample-size implication, which governs §10. The standard error of an annualised Sharpe estimate over  $T$  years is approximately  $\sqrt{(1 + S^2/2)/T}$ :

| Sample   | $S = 0.4$             | $S = 1.0$             |
|----------|-----------------------|-----------------------|
| 5 years  | se 0.46 ( $t = 0.9$ ) | se 0.55 ( $t = 1.8$ ) |
| 10 years | se 0.33 ( $t = 1.2$ ) | se 0.39 ( $t = 2.6$ ) |
| 20 years | se 0.23 ( $t = 1.7$ ) | se 0.27 ( $t = 3.7$ ) |
| 50 years | se 0.15 ( $t = 2.7$ ) | se 0.17 ( $t = 5.8$ ) |

A twenty-year backtest cannot distinguish a Sharpe of 0.4 from zero at conventional significance. This is why the century-long studies of §4.2 exist, and why the arguments about them are so hard to settle.

## §5 Key takeaways

1. Expected P&L is  $\sum_k w_k \gamma_{k+1} + \mu^2 \sum_k w_k$  — a weighted sum of return autocovariances plus a drift term. Trend-following is long autocovariance and long squared drift.
2. The kernel must have mass where the autocorrelation does. A slow filter on fast-decaying autocorrelation earns exactly nothing, however well built.
3. The drift term biases the standard test positive under the null. Demean each market before testing; treat undemeaned trend results as uninformative.
4. Exact identity: the price-versus-moving-average rule with window  $n$  earns  $\frac{1}{2}\gamma_0(\text{VR}(n) - 1)$  per bar. The trend rule and the variance-ratio test are the same statistic; estimating the VR profile tells you whether to bother backtesting.
5. Exact identity, no assumptions: realised P&L equals a convexity boundary term, plus accumulated trend energy  $\sum x_t^2$ , minus accumulated realised variance  $\sum r_t^2$ . You are long trend vol and short realised vol.
6. The strategy is a small residual between two large terms, which is the structural reason it is so cost- and parameter-sensitive.
7. The straddle analogy is exact for smooth paths and wrong for gaps: the convexity is synthetic and requires a tradeable market to replicate.
8. The optimal lookback is set by trend persistence  $\tau$ , not by signal strength — span  $\approx 2\tau$ . And the optimum is flat: a factor-of-two error costs about 5%.
9. Multi-timescale ensembles buy insurance against mis-specification, not return. In simulation they never beat the best single span.
10. Calibration to memorise: a diversified Sharpe near 1.0 corresponds to a daily autocorrelation of about 0.005, undetectable in one market ( $t \approx 0.25$  over a decade). Diversification is the mechanism, and lowering  $\bar{\rho}$  beats raising  $N$ .

## 6. Trend estimation and detection

This is the catalogue. Every method gets the same fields in the same order, so they can be compared directly: intuition, definition, kernel, assumptions, strengths, weaknesses, cost, failure modes, when preferred. Where a method is a linear filter I give its kernel explicitly, because §7 shows that the kernel is what actually distinguishes them.

A note before starting. Reading a catalogue like this creates a strong impression that these are ten different strategies. They are not. Six of the ten are the same linear filter with different weights, and §5.8 showed that the weights matter far less than the timescale. Read this section for the failure modes, which genuinely differ, more than for the definitions, which mostly do not.

### 6.1 Lookback return (time-series momentum)

Intuition. Is the price higher than it was  $L$  bars ago? If yes, be long.

Definition.  $s_t = r_{t-L:t} = p_t - p_{t-L}$ , with position  $\pi_t = g(s_t)$ ; the canonical academic form takes  $g = \text{sign}$  and  $L = 252$ .

Kernel.  $w_j = \mathbb{1}\{j < L\}$  — rectangular. Every return in the window counts equally; nothing outside counts at all.

Assumptions. That the sum of returns over exactly  $L$  bars is informative about the next one, with no preference for recency.

Strengths. Maximally simple, no parameters beyond  $L$ , and it is the form in which almost all academic evidence is stated, so it is the right choice when you want to compare against the literature.

Weaknesses. The rectangular kernel gives the return from  $L$  bars ago exactly the same weight as yesterday's, which is implausible under any diffusion story, and gives a return from  $L + 1$  bars ago zero weight — a discontinuity with no economic meaning. The signal therefore jumps when a large old return drops out of the window, an artefact with no information content.

Cost.  $O(1)$  per bar from the running sum.

Failure modes. (i) The drop-out artefact above generates trades on no news. (ii) It is maximally exposed to the drift confound of §5.3, because  $\text{sign}(p_t - p_{t-L})$  is positive most of the time in a drifting market. (iii) With  $g = \text{sign}$  it discards the signal magnitude, losing the convexity of §1.5.

When preferred. Benchmarking, and communication. I would not run it in production.

## 6.2 Price versus moving average

Intuition. Is the price above its own recent average?

Definition.  $s_t = p_t - \text{MA}_n(p)_t$ , where MA is a simple or exponential average over  $n$  bars.

Kernel.  $w_j = (n - 1 - j)^+ / n$  for the simple average — a descending ramp from  $(n-1)/n$  at lag 0 down to zero at lag  $n - 1$ . For the exponential average,  $w_j = (1 - \alpha)^{j+1}$  — a clean exponential decay. Both derived in §5.4 and §7.2.

Assumptions. That recent returns matter more than old ones, linearly (SMA) or geometrically (EWMA).

Strengths. The decay is economically sensible, there is no drop-out artefact for the EWMA version, and — by §5.4 — its expected P&L is exactly  $\frac{1}{2}\gamma_0(\text{VR}(n) - 1)$ , which makes it the most analytically tractable rule in the catalogue.

Weaknesses. Maximum weight sits on the most recent return, which is the noisiest one and the one most contaminated by microstructure effects (bid-ask bounce, closing auctions). This makes the raw signal jumpy and turnover high.

Cost.  $O(1)$  per bar (EWMA),  $O(1)$  with a running sum (SMA).

Failure modes. (i) In a fast-mean-reverting market the heavy weight on lag 0 makes it a reversal signal with the wrong sign. (ii) It has no notion of scale, so the raw  $s_t$  is not comparable across markets or across volatility regimes — you must normalise (§8.3), and forgetting to is the most common implementation bug in this whole catalogue.

When preferred. As the default. If someone asks for one trend rule, this is it, in its EWMA form, normalised by volatility.

## 6.3 Moving-average crossover

Intuition. Is the short-term average above the long-term one? Equivalently: is the recent past better than the more distant past?

Definition.  $s_t = \text{MA}_m(p)_t - \text{MA}_n(p)_t$  with  $m < n$ ; the EWMA version (“EWMAC”) with spans  $m$  and  $n$  is the industry standard.

Kernel.  $w_j = \frac{(n - 1 - j)^+}{n} - \frac{(m - 1 - j)^+}{m}$ , which — as §7.2 shows — rises linearly to a peak at lag  $m - 1$  and then decays linearly to zero at lag  $n - 1$ . A band-pass filter: it deliberately down-weights the most recent returns.

Assumptions. That the informative frequency band is bounded away from both zero and the Nyquist frequency: neither the very recent (noise) nor the very distant (stale) past is what you want.

Strengths. The down-weighting of recent returns removes exactly the microstructure noise that hurts §6.2, giving materially lower turnover for similar gross edge. This is why it, and not the simpler rule, is what most production systems run.

Weaknesses. Two parameters instead of one, and their ratio interacts with the timescale in a way that is easy to over-fit. The lag introduced by suppressing recent returns costs you at turning points.

Cost.  $O(1)$  per bar.

Failure modes. (i) Certain  $(m, n)$  combinations produce kernels with a negative lobe — an implicit mean-reversion component the designer did not intend. Beekhuizen and Hallerbach (2017) document this for common multi-average rules; it is invisible in price space and obvious in return space. (ii) The signal crosses zero frequently in choppy markets, which with a sign response generates a burst of loss-making trades — the classic whipsaw.

When preferred. Production, when turnover matters. Use spans in roughly a 1:4 ratio and set the slow span from  $\tau$  per §5.8.

## 6.4 Regression slope

Intuition. Fit a straight line to the recent log price and use its slope.

Definition.  $\hat{\beta}_t = \arg \min_{\beta} \sum_{i=0}^{L-1} (p_{t-i} - a - \beta(t-i))^2$ . A common variant divides by the standard error of the slope, giving a  $t$ -statistic, which normalises for volatility automatically.

Kernel.  $w_j = \frac{6(j+1)(L-1-j)}{L(L^2-1)}$  — a centred inverted parabola, symmetric, peaking in the middle of the window and vanishing at both ends. (I verified this closed form against direct computation; the weights sum to one, so a pure straight-line path returns its own slope.)

Assumptions. That the log price is locally linear plus independent noise, and that the noise is homoskedastic within the window.

Strengths. The  $t$ -statistic variant is self-normalising, which is elegant. The symmetric kernel is maximally smooth, giving a very stable signal.

Weaknesses. The symmetry is also the problem: weighting the middle of the window most means the estimate lags by  $\approx L/2$  bars. This is the most lagged rule in the catalogue. Its noise-rejection is bought entirely with timeliness.

Cost.  $O(1)$  per bar with running sums of  $p$ ,  $tp$ ,  $t$ ,  $t^2$ .

Failure modes. (i) The lag makes it late to reversals, so it gives back more at turning points than the exponential rules. (ii) The  $t$ -statistic version divides by an in-window residual standard error, which is a very noisy volatility estimate for short  $L$  and produces enormous signal spikes when a window happens to be quiet. Use a separate, longer volatility estimate instead.

When preferred. When signal stability matters more than responsiveness, and in research settings where the  $t$ -statistic's interpretability helps.

## 6.5 Breakout and channel rules

Intuition. Has the price exceeded its highest level of the last  $L$  bars?

Definition. Donchian: go long when  $P_t > \max_{1 \leq i \leq L} P_{t-i}$ , short when  $P_t < \min_{1 \leq i \leq L} P_{t-i}$ , hold otherwise. Variants add a shorter exit channel or a volatility-scaled band around a moving average (Keltner, Bollinger).

Kernel. None — this is not a linear filter. The signal is a function of order statistics of the price path, so §5.2's machinery does not apply and its expected P&L cannot be written as a weighted sum of autocovariances.

Assumptions. That extremes are informative beyond what the mean of the window conveys — i.e. that the maximum carries information the average does not.

Strengths. Naturally discrete: it produces few, well-separated trades and therefore low turnover. It is insensitive to the shape of the return distribution within the channel. And it has an interpretable economic story (resting stop orders clustered above the range).

Weaknesses. It throws away almost all the data — only the extremes matter — and it is discontinuous, so a one-tick difference flips the position.

Cost.  $O(\log L)$  per bar with a monotonic deque; naively  $O(L)$ .

Failure modes. (i) The extreme of a window is a maximally noisy statistic; a single bad print or a stale quote creates a false breakout, so data cleaning matters more here than for any other rule. (ii) In markets with limit moves or frequent gaps, the breakout level is often jumped rather than touched, so the realised entry differs systematically from the backtested one — this rule's backtests are optimistic in exactly the markets where trend-following makes its money. (iii) Being flat inside the channel means the position is a step function of the price, which makes portfolio-level risk lumpy.

When preferred. When costs dominate and you need a genuinely low-turnover system, or as a diversifier alongside a linear filter, since its errors are uncorrelated with theirs.

## 6.6 State-space models and the Kalman filter

Intuition. Posit that there is an unobserved trend, and estimate it optimally.

Definition. The local level model of §5.8, or its local linear trend extension in which both a level and a slope evolve:

$$\begin{aligned} p_t &= \ell_t + \varepsilon_t \\ \ell_t &= \ell_{t-1} + b_{t-1} + \xi_t \\ b_t &= \phi b_{t-1} + \eta_t \end{aligned}$$

with  $b_t$  the trend. The Kalman filter delivers  $\mathbb{E}[b_t \mid \mathcal{F}_t]$  and, importantly, its variance.

Kernel. In steady state, an EWMA with  $\alpha^* = 1 - \phi(1 - K)$  (§5.8) — so this is not a new estimator, it is the exponential rule with a principled parameter. That equivalence is the most useful thing in this subsection.

Assumptions. Linear Gaussian dynamics with known variances. The Gaussian part is badly violated by financial returns; the linearity is fine.

Strengths. It tells you the uncertainty of the trend estimate, not just its value, which nothing else here does. That enables principled position sizing (scale by  $\hat{b}_t / \text{sd}(\hat{b}_t)$ ) and honest confidence intervals. It also handles missing data and irregular sampling natively, which matters for a global futures panel with non-overlapping holidays.

Weaknesses. Estimating  $\sigma_\eta^2$  and  $\sigma_\varepsilon^2$  from data is a badly conditioned problem at realistic signal-to-noise ratios — you are trying to estimate a variance ratio of order  $10^{-3}$  — and maximum-likelihood estimates are unstable. In practice people fix the parameters by judgement, which forfeits most of the method's claimed advantage.

Cost.  $O(d^2)$  per bar for a  $d$ -dimensional state; trivially cheap here.

Failure modes. (i) Fitting the variances by MLE on a short sample typically collapses to a corner solution — either  $\hat{\sigma}_\eta \approx 0$  (no trend ever) or a very fast filter. (ii) The Gaussian assumption means a single large return is treated as strong evidence of a trend change rather than as a fat-tail draw, so the filter over-reacts to jumps. A Student- $t$  observation model or simple winsorisation fixes this and is worth the trouble.

When preferred. When you need uncertainty estimates, when data is irregular, or when you want to justify a lookback from a model rather than a backtest.

## 6.7 Smoothers: Hodrick–Prescott, $\ell_1$ , and wavelets

Intuition. Separate the price into a smooth component and a rough one, and trade the smooth one's direction.

Definition. Each solves a penalised least-squares problem over the whole path. Hodrick–Prescott minimises  $\sum_t (p_t - \tau_t)^2 + \lambda \sum_t ((\tau_{t+1} - \tau_t) - (\tau_t - \tau_{t-1}))^2$ , giving a smooth trend;  $\ell_1$  trend filtering (Kim, Koh, Boyd and Gorinevsky, 2009) replaces the squared second-difference penalty with an absolute one, giving a piecewise linear trend with a small number of kinks — a principled formalisation of “the trend changed here”. Wavelet methods decompose the path into scales and reconstruct from the coarse ones.

Kernel. For HP, a symmetric two-sided kernel — which is the crux.

Assumptions. That a smoothness penalty encodes something true about the price-generating process.

Strengths.  $\ell_1$  trend filtering in particular produces exactly the object practitioners describe informally — straight-line trend segments joined at breakpoints — with a convex, globally solvable formulation and one interpretable parameter.

Weaknesses. These are smoothers, not filters, and the distinction is the whole story. They use future data to estimate the trend at time  $t$ . Their excellent in-sample fit is not available in real time.

Cost. HP is  $O(T)$  for the whole path via a banded solve;  $\ell_1$  is a convex program,  $O(T)$  per interior-point iteration.

Failure modes. (i) The look-ahead trap. Applying HP or  $\ell_1$  to the full sample and then backtesting on the extracted trend produces spectacular, entirely fictitious results. This is the single most common serious error I have seen in trend research. The only correct use is to re-solve on the expanding window  $\{p_1, \dots, p_t\}$  at every  $t$ , which is expensive and which degrades the estimate exactly at the right-hand edge where you need it. (ii) Even used correctly, the endpoint estimate of a two-sided smoother is far noisier than the interior, so its apparent smoothness is misleading.

When preferred. For analysis — labelling historical regimes, defining trend episodes for study, generating targets for supervised learning. Rarely for live signal generation, where a one-sided filter is the honest choice.

## 6.8 Change-point detection

Intuition. Rather than estimating a trend continuously, detect the moments when the regime changes.

Definition. The classical CUSUM statistic accumulates deviations from a reference and signals when the accumulation exceeds a threshold:  $S_t = \max(0, S_{t-1} + r_t - k)$ , alarm when  $S_t > h$ . Bayesian online change-point detection maintains a posterior over the time since the last change.

Kernel. Nonlinear and adaptive: the effective lookback resets at each detection, which is exactly the property a fixed-kernel filter lacks.

Assumptions. That regimes are genuinely discrete, and that the pre- and post-change distributions are estimable.

Strengths. Optimal detection delay for a given false-alarm rate (this is CUSUM's classical guarantee), and an adaptive look-back for free. Combining change-point detection with a trend filter is one of the more promising current research directions — the changepoint tells you when to forget.

Weaknesses. The pre/post distributions are not known and must be estimated from very few observations near a change. And the “regimes are discrete” assumption is a modelling convenience, not a fact about markets.

Cost.  $O(1)$  per bar for CUSUM,  $O(T)$  per bar naively for Bayesian online detection,  $O(1)$  amortised with pruning.

Failure modes. (i) The threshold  $h$  trades false alarms against detection delay, and at trend-following's signal-to-noise ratios (§5.9) the operating point is grim: to get a tolerable false-alarm rate you accept a delay comparable to the trend's own duration. (ii) Detections cluster during volatile periods, producing exactly the behaviour you do not want — maximum signal churn when markets are most expensive to trade.

When preferred. As a modifier on a filter — shortening the effective lookback after a detection — rather than as a standalone signal.

## 6.9 Persistence statistics

Intuition. Rather than estimate the direction of a trend, estimate whether this market is the kind of market that trends at all.

Definition. Three in common use:

- Variance ratio  $VR(q)$  as defined in §5.4 — above one means trending. By §5.4 this is not merely diagnostic but is the expected P&L.
- Hurst exponent  $H$ , from the scaling  $\text{Var}(p_{t+q} - p_t) \propto q^{2H}$ .  $H = 1/2$  is a random walk;  $H > 1/2$  is persistence. Note  $VR(q) \propto q^{2H-1}$ , so  $H$  and the VR profile's slope carry the same information.
- Efficiency ratio (Kaufman):  $|p_t - p_{t-L}| / \sum_{i=0}^{L-1} |r_{t-i}|$  — net progress divided by gross path length, in  $[0, 1]$ . A pure trend gives 1; a random walk gives roughly  $\sqrt{2/(\pi L)} \cdot \dots$ , i.e. something small.

Assumptions. Stationarity over the estimation window — which is a strong assumption for a statistic whose whole purpose is to detect regime differences.

Strengths. These are conditioning variables, not signals: they tell you how much to trust a trend signal, or which markets to include, or when to reduce size. Used that way they are among the most defensible tools here.

Weaknesses. All three are noisy.  $H$  in particular requires far more data than people use, and the standard estimators (rescaled range in particular) have severe small-sample bias — a random walk of a few hundred points routinely yields  $\hat{H} \approx 0.6$ .

Cost.  $O(L)$  to  $O(L \log L)$  per update.

Failure modes. (i) Estimating  $H$  or VR on short windows and treating the result as a regime classification. The estimator's own noise dominates. (ii) Circularity: selecting markets by their historical variance ratio and then reporting the trend performance of the selected set is straightforward selection bias, and it is common.

When preferred. For market selection at long horizons and for research diagnostics. Not as a real-time regime switch.

## 6.10 Machine-learned signals

Intuition. Learn the map from past returns to position directly, rather than specifying it.

Definition. Lim, Zohren and Roberts (2019) train an LSTM to output a position directly by maximising a Sharpe-ratio objective, using volatility-normalised returns and a set of standard trend signals as inputs; subsequent work replaces the recurrent architecture with attention and adds change-point features.

Kernel. Learned, nonlinear, and time-varying — which is the whole proposition and the whole risk.

Assumptions. That there is enough signal, relative to the number of effective parameters, to identify a richer function than a linear filter.

Strengths. It can learn the response function  $g$  and the kernel jointly, and it can condition on state (volatility, correlation, positioning). The reported improvements over a classical rule are real in the papers.

Weaknesses. The signal-to-noise calibration of §5.9 is the problem. With a per-market daily information coefficient of order 0.01 and a few thousand effectively independent observations, the data supports estimating a handful of parameters, not thousands. [Contested] — the published improvements are genuine on their samples; whether they survive as out-of-sample premia is not established.

Cost. Training cost is irrelevant; the relevant cost is the number of research trials, which is what §10.5 charges you for.

Failure modes. (i) Learning the sample's specific volatility regime rather than a trend relationship — the models are typically trained on volatility-scaled returns precisely to prevent this, and it is worth verifying it worked. (ii) Silent look-ahead through the normalisation: computing the volatility scaling or the feature standardisation on the full sample leaks the future, and in a low-signal setting the leak is bigger than the signal.

When preferred. When you have a genuinely large cross-section, a disciplined out-of-sample protocol, and a specific hypothesis about what nonlinearity or conditioning you expect to find. Not as a first system.

## 6.11 Comparison

Ratings are mine, on the calibration of §5.9 rather than on any single dataset.

| Method                  | Linear? | Params | Turnover | Lag      | Robustness | Main failure mode                 |
|-------------------------|---------|--------|----------|----------|------------|-----------------------------------|
| Lookback return         | Yes     | 1      | Medium   | Medium   | Medium     | Drift confound; window drop-out   |
| Price vs MA             | Yes     | 1      | High     | Low      | Medium     | Overweights noisiest lag          |
| MA crossover            | Yes     | 2      | Medium   | Medium   | Good       | Unintended negative kernel lobe   |
| Regression slope        | Yes     | 1      | Low      | High     | Good       | Late at turning points            |
| Breakout                | No      | 1–2    | Low      | Medium   | Medium     | Gap-through; optimistic backtests |
| Kalman / state space    | Yes†    | 2–3    | Medium   | Low      | Medium     | Unstable variance estimation      |
| HP / $\ell_1$ smoothers | Yes‡    | 1      | Low      | —        | Poor§      | Look-ahead if misused             |
| Change-point            | No      | 2      | Bursty   | Low      | Poor       | Bad delay/false-alarm trade-off   |
| Persistence statistics  | No      | 1–2    | —        | —        | Poor       | Estimator noise; selection bias   |
| Machine-learned         | No      | Many   | Variable | Variable | Unknown    | Over-parameterisation; leakage    |

† Linear in steady state — it is an EWMA (§5.8). ‡ Linear but two-sided, hence not a causal filter. § Robustness refers to real-time use; as offline analysis tools these are fine.

## §6 Key takeaways

1. Six of these ten are the same linear filter with different weights. Read the catalogue for failure modes, not for definitions.
2. Price versus an exponential moving average, volatility-normalised, is the right default. Everything else needs a reason.
3. The crossover's real advantage is that it down-weights the most recent returns, cutting microstructure noise and turnover — not that it is a “faster/slower” comparison.
4. The regression slope's symmetric kernel buys smoothness with a lag of about half the window. That trade is sometimes right and is always explicit.
5. Breakouts are not linear filters and cannot be analysed with §5's machinery. Their backtests are systematically optimistic in gapping markets.
6. The Kalman filter is an EWMA with a principled decay. Its real contribution is the uncertainty estimate, which enables honest sizing.
7. HP and  $\ell_1$  smoothers are two-sided. Using them without re-solving on an expanding window is the most common serious look-ahead error in trend research.
8. Persistence statistics belong in market selection and diagnostics, not in real-time switching — their estimator noise exceeds the effect at any window short enough to be responsive.

## 7. Taxonomy and equivalences

### 7.1 The master form, with slots

The pipeline of §1.3, written as a design space rather than a formula:

$$\pi_{i,t} = \underbrace{c_t}_{\text{portfolio}} \cdot \underbrace{\frac{\sigma^*}{\hat{\sigma}_{i,t}}}_{\text{risk}} \cdot \underbrace{g}_{\text{response}} \left( \underbrace{\sum_k w_k \tilde{r}_{i,t-k}}_{\text{kernel}} \right)$$

| Slot                 | What it controls                                               | Choices                                                       |
|----------------------|----------------------------------------------------------------|---------------------------------------------------------------|
| $\{w_k\}$            | Which horizon you are betting on, and how the past is weighted | box, ramp, exponential, hump, parabola, learned               |
| $g(\cdot)$           | The shape of the payoff and the turnover                       | linear, capped-linear, sign, bump-shaped, thresholded         |
| $\hat{\sigma}_{i,t}$ | Risk comparability across markets and time                     | EWMA of squared returns, range estimators, blends of horizons |
| $c_t$                | Total portfolio risk and its dynamics                          | fixed target, correlation-adjusted, drawdown-modulated        |

The value of writing it this way is that the slots are close to orthogonal. You can change the kernel without touching the response; you can change the volatility estimator without touching either. That means you can reason about which slot to spend effort on, which §7.3 does.

### 7.2 The kernel table: exact equivalences

Every linear rule in §6 is a choice of  $\{w_k\}$ , and the weights have closed forms. This table is, I think, the single most useful half-page here, because it collapses a folklore of “indicators” into one object.

| Rule                                        | Weight $w_j$ on $r_{t-j}$                                    | Shape            | Peak at lag                                                  | Centre of mass            |
|---------------------------------------------|--------------------------------------------------------------|------------------|--------------------------------------------------------------|---------------------------|
| $L$ -bar lookback return<br>$p_t - p_{t-L}$ | $\mathbb{1}\{j < L\}$                                        | box              | flat                                                         | $(L - 1)/2$               |
| Price — SMA( $n$ )                          | $\frac{(n - 1 - j)^+}{n}$                                    | descending ramp  | 0                                                            | $(n - 2)/3$               |
| Price — EWMA( $\alpha$ )                    | $(1 - \alpha)^{j+1}$                                         | exponential      | 0                                                            | $(1 - \alpha)/\alpha$     |
| SMA( $m$ ) — SMA( $n$ ),<br>$m < n$         | $\frac{(n - 1 - j)^+}{m} - \frac{(m - 1 - j)^+}{m}$          | tent / band-pass | $m - 1$                                                      | —                         |
| EWMAC( $\alpha_f, \alpha_s$ )               | $(1 - \alpha_s)^{j+1} - (1 - \alpha_f)^{j+1}$                | hump / band-pass | $\approx \frac{\ln(\alpha_f/\alpha_s)}{\alpha_f - \alpha_s}$ | —                         |
| OLS slope over $L$<br>bars                  | $\frac{6(j + 1)(L - 1 - j)}{L(L^2 - 1)}$                     | centred parabola | $(L - 2)/2$                                                  | $(L - 2)/2$               |
| Kalman, local level,<br>steady state        | $\propto (1 - \alpha^*)^j$ ,<br>$\alpha^* = 1 - \phi(1 - K)$ | exponential      | 0                                                            | $(1 - \alpha^*)/\alpha^*$ |

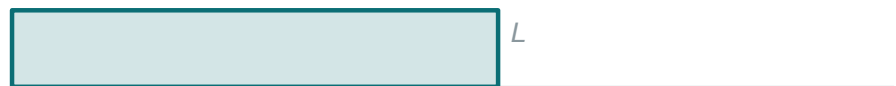
Every row is an algebraic identity, not an approximation, and I verified each numerically against direct computation on simulated paths. The derivations for the ramp and the parabola are in §5.4 and §6.4; the exponential and the two band-pass rows follow the same telescoping argument.

*Every measure is a weighted sum of past returns. Only the kernel differs.*

## Lookback return

*rectangular*

$w_k = 1/L, k < L$



## EWMA

*exponential*

$w_k \propto \lambda^k$



## MA displacement

*descending ramp*

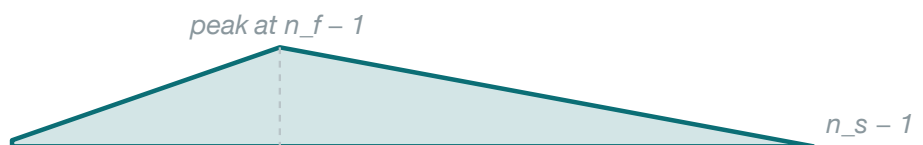
$w_k \propto n-1-k$



## MA crossover

*hump / band-pass*

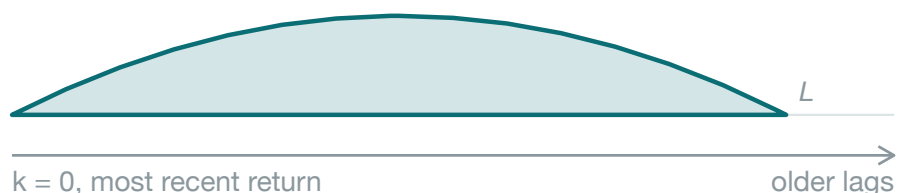
*fast — slow*



## Regression slope

*centred parabola*

$w_k \propto (k+1)(L-1-k)$



Four readings that change how you design a system:

A moving-average crossover down-weights recent returns. The hump peaks at an interior lag, so yesterday's return — the noisiest observation you have — gets less weight than the return from  $m$  bars ago. That is a feature, and it is the actual

reason crossovers outperform the simpler rule on a net-of-cost basis. Most descriptions of the crossover (“fast average versus slow average”) obscure this completely.

Some popular rules contain hidden mean reversion. If you build a signal from several moving averages with badly chosen windows, the resulting kernel can go negative at some lags — meaning the rule bets against returns from that part of the past. Beekhuizen and Hallerbach (2017) document this for common multi-average constructions. It is invisible in price space. The remedy is one line of code: expand your rule into return weights and plot them. If any  $w_j < 0$  and you did not intend it, fix the rule.

“Which lookback?” has a precise meaning. Two rules with different names and similar centres of mass are near-identical strategies; two rules with the same name and different windows are not. Compare rules by their kernels’ centres of mass, not by their parameter labels. A price-versus-SMA(200) rule has its centre of mass at  $(200 - 2)/3 \approx 66$  bars, while a price-versus-EWMA rule of span 200 has it at  $(1 - \alpha)/\alpha \approx 100$  bars. Same nominal “200-day rule”, 50% different effective horizon.

The regression slope is the most lagged rule you can build. Its centre of mass is at  $(L - 2)/2$ , versus  $(n - 2)/3$  for the moving-average rule — for the same nominal window it looks 50% further into the past.

### 7.3 Which slot matters

This is the payoff of building the taxonomy. Ranked by the size of the effect I observe in simulation and by what the literature supports:

| Rank | Slot                              | Effect size                                                                      | Evidence                                                                                                                   |
|------|-----------------------------------|----------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------|
| 1    | Risk scaling $\hat{\sigma}_{i,t}$ | Sharpe 0.04 $\rightarrow$ 0.36 in my simulation with volatility clustering       | Kim, Tse & Wald (2016) attribute much of the reported premium to it; Harvey et al. (2018) document its effect on drawdowns |
| 2    | Portfolio aggregation $c_t$       | Sharpe $\times 2.45$ for 50 markets at $\bar{\rho} = 0.15$ (\$5.9)               | Arithmetic; Baltas & Kosowski (2013) on correlation adjustment                                                             |
| 3    | Kernel timescale                  | $\pm \$5\%$ for a factor-of-two error, $-\$50\%$ for a factor of sixteen (\$5.8) | My simulation; consistent with the flatness reported across the literature                                                 |
| 4    | Response function $g$             | Changes skew and hit rate substantially; changes Sharpe modestly                 | \$1.5; Lempérière et al. (2014) find saturation for large signals                                                          |
| 5    | Kernel shape at fixed timescale   | Small                                                                            | Levine & Pedersen (2016); Beekhuizen & Hallerbach (2017)                                                                   |

The ordering is close to the inverse of the attention these get in the practitioner literature, which is dominated by discussion of indicators (rank 5) and lookbacks (rank 3). If you have one week to improve a trend system, spend it on the volatility estimator and the correlation structure.

One caveat on rank 1 that is easy to miss. The reason volatility scaling helps so much is not that it improves the signal — it does not. It is that (i) it makes risk comparable across markets so the portfolio is not dominated by whichever instrument is currently most volatile, and (ii) volatility is far more predictable than returns, so dividing by a forecastable quantity sharpens the ratio. [Contested] Kim, Tse and Wald (2016) push this further and argue that much of the reported time-series momentum premium is the volatility scaling rather than the trend signal. If they are right, a chunk of what the industry calls trend-following is really a volatility-managed long position, which would be a different product with a different fee.

### 7.4 Same thing, different names; same name, different things

| Claim                                                                                              | Status                                                                                                                               |
|----------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------|
| Price-vs-MA( $n$ ) rule $\equiv$ averaging $L$ -bar momentum over $L = 1..n - 1$                   | Exact (§5.4)                                                                                                                         |
| Its expected P&L $\equiv \frac{1}{2}\gamma_0(\text{VR}(n) - 1)$                                    | Exact (§5.4)                                                                                                                         |
| Kalman filter on the local level model $\equiv$ EWMA with $\alpha^* = 1 - \phi(1 - K)$             | Exact in steady state (§5.8)                                                                                                         |
| Hurst exponent $H \equiv$ slope of $\log \text{VR}(q)$ vs $\log q$ , shifted<br>MACD $\equiv$ EWMA | Exact given the scaling assumption ( $\text{VR}(q) \propto q^{2H-1}$ )<br>Exact — MACD is an EWMA crossover, usually spans 12 and 26 |
| “Trend-following” $\equiv$ “time-series momentum”                                                  | Approximate. TSMOM is the sign-response, box-kernel special case                                                                     |
| “Trend-following” $\equiv$ “managed futures”                                                       | False. One is a strategy, the other an industry with fees                                                                            |
| “Momentum” (cross-sectional) $\equiv$ “momentum” (time-series)                                     | False and consequential. Different signals, different portfolios, different risks — see Goyal and Jegadeesh’s decomposition          |
| Bollinger-band breakout $\equiv$ Donchian breakout                                                 | False. One is a volatility-scaled band around a mean, the other an order statistic                                                   |
| Volatility targeting $\equiv$ risk parity                                                          | False. The first sets total risk over time, the second allocates risk across assets                                                  |

The row worth pausing on is cross-sectional versus time-series momentum. They share a word and are routinely conflated, but a cross-sectional momentum portfolio is dollar-neutral by construction while a time-series momentum portfolio has a net directional position that varies with how many markets are trending. That net position is where most of trend-following’s crisis behaviour comes from, and it is exactly what the cross-sectional version lacks.

## 7.5 The design space



The two teal boxes are where the return is. The two grey boxes are where the literature is.

## §7 Key takeaways

1. Expand every rule into its return-space kernel. It is a few lines of code and it collapses a folklore of indicators into one comparable object.
2. The kernel table entries are exact identities, not analogies — including the equality of the moving-average rule's expected P&L with half the variance-ratio excess.
3. A crossover's advantage is that it down-weights the most recent return. Describing it as “fast versus slow” hides the only thing that matters.
4. Check your kernel for unintended negative lobes. Some popular multi-average rules contain hidden mean reversion.
5. Compare rules by their kernels' centre of mass, not their parameter names: SMA(200) and EWMA(span 200) differ by 50% in effective horizon.
6. Effort ordering: volatility scaling, then correlation structure, then timescale, then response, then kernel shape. This inverts the attention these receive in practitioner writing.
7. Cross-sectional and time-series momentum share a name and are different strategies. The net directional exposure of the time-series version is the source of most of its distinctive behaviour.

## 8. From signal to portfolio

### 8.1 Effort ordering

Before any of the detail, the ordering — because the single most common way to waste six months on a trend system is to spend them on the indicator.

1. Data correctness. Not a component, a precondition. A single mis-stitched futures roll can create or destroy an entire market's apparent edge (§8.2). This is worth more effort than everything below it combined.
2. Volatility estimation and scaling (§8.3). Rank 1 of §7.3.
3. Universe construction (§8.6). Which markets, and how correlated.
4. Portfolio risk aggregation (§8.6).
5. Kernel timescale (§5.8).
6. Cost model and trading rate (§8.7, §8.8).
7. Response function (§8.4).
8. Kernel shape. Last.

A blunt way to put it: a mediocre indicator on clean data with correct sizing across 60 markets beats a superb indicator on dirty data across 5.

### 8.2 The data layer

Trend-following lives in futures, and futures data has three problems that do not exist in equities.

Contract stitching. A futures “price series” is a fiction — it is a sequence of distinct contracts glued together. At each roll the front and next contracts trade at different prices, and the gap is not a return. Three conventions:

| Method                 | What it does                            | Use for                                                                            |
|------------------------|-----------------------------------------|------------------------------------------------------------------------------------|
| Unadjusted             | Concatenate raw prices                  | Nothing. It injects a fake return at every roll                                    |
| Back-adjusted (panama) | Shift the whole history by the roll gap | Signal generation. Returns are correct; levels are meaningless and can go negative |

| Method          | What it does                                         | Use for                                                      |
|-----------------|------------------------------------------------------|--------------------------------------------------------------|
| Ratio-adjusted  | Multiply history by the price ratio                  | Signal generation when you need positive prices              |
| Return-stitched | Compute returns within contract, concatenate returns | The cleanest. Reconstruct a level series if a rule needs one |

[Practice] I would use return-stitching and rebuild a synthetic level series from the returns. It makes the roll handling explicit rather than implicit, and it removes the temptation to use price levels in a rule — which back-adjusted series silently corrupt. Note the trap in back-adjusted data specifically: a breakout rule computed on a back-adjusted series is comparing today's price to a historical extreme that has been shifted by the accumulated roll gaps, which is not the extreme that market participants saw.

Roll timing and look-ahead. Rolling on a fixed calendar day is reproducible; rolling on open-interest crossover is realistic but requires care, because open-interest data is often published with a lag. Using it same-day is look-ahead. [Practice] Roll on a fixed schedule, a few days before first notice, and lag any volume- or open-interest-based decision by at least one bar.

Non-synchronous closes. A global futures panel has markets closing across 18 hours. Correlations computed on same-labelled-date closes are biased downward for markets in distant time zones, and a signal computed at one market's close may use another's stale price. [Practice] Either sample all markets at a common wall-clock time, or accept the bias and use overlapping multi-day returns for correlation estimation, which reduces it.

Survivorship. Delisted contracts, failed currencies, and markets that stopped trading must be in the sample. A universe of markets that still trade in 2026 is a universe selected for having survived, and trend-following's tail is precisely where the survivors and non-survivors differ.

### 8.3 Volatility estimation

The most important component. What you need is a forecast of the next bar's return volatility, from information available now.

The standard choice is an EWMA of squared returns,  $\hat{\sigma}_t^2 = (1 - \lambda)\hat{\sigma}_{t-1}^2 + \lambda r_t^2$ , with  $\lambda$  corresponding to a span of roughly 20–60 bars. Choices worth making deliberately:

- Speed. A fast estimator tracks regime changes but injects its own noise into the position and therefore into turnover. A slow one is stable but leaves you over-sized entering a volatility spike. [Practice] Blend: use  $\hat{\sigma}_t = \frac{1}{2}\hat{\sigma}_t^{\text{fast}} + \frac{1}{2}\hat{\sigma}_t^{\text{slow}}$ , or equivalently a mixture of a 20-day and a 6-month estimate. The blend is Carver's recommendation and it is meaningfully better than either alone, because the slow component anchors the estimate during transient spikes.
- Range estimators. Parkinson and Garman–Klass estimators use the bar's high and low, and are several times more efficient than close-to-close for the same window. Baltas and Kosowski (2013) find that more efficient volatility estimation cuts portfolio turnover by more than a third without significant performance loss — which, given \$8.8, is a large effect. Their weakness is sensitivity to bad prints in the high and low, so winsorise.
- The floor. A volatility estimate can get arbitrarily small in a quiet market, and  $\sigma^*/\hat{\sigma}_t$  then explodes. Always floor the estimate, at something like the 10th percentile of its own history, and cap the resulting position independently.

Do not compute the volatility scaling with full-sample data. This is the most common look-ahead in trend backtests and it is very hard to see, because the resulting equity curve looks plausible rather than absurd. Every  $\hat{\sigma}_t$  must use only  $\{r_s : s \leq t\}$ .

How much does this matter? In my simulation with realistic volatility clustering, holding the signal identical:

| Position rule                     | Sharpe |
|-----------------------------------|--------|
| Raw signal $x_t$ , no scaling     | 0.04   |
| $x_t/\hat{\sigma}_t$              | 0.21   |
| $\text{sign}(x_t)$                | 0.12   |
| $\text{sign}(x_t)/\hat{\sigma}_t$ | 0.36   |

Note that  $\text{sign}(x_t)/\hat{\sigma}_t$  beat  $x_t/\hat{\sigma}_t$ . The reason is instructive: the raw signal's magnitude is itself contaminated by volatility — a big  $x_t$  may mean a strong trend or merely a volatile market — so dividing a slow trend estimate by a fast volatility estimate leaves noise in the ratio. Taking the sign separates the two cleanly: direction from the slow filter, size from the volatility estimate. If you want a magnitude-sensitive response (and §1.5 says you do, for the convexity), normalise the signal by its own running scale first, then apply the response, then scale by volatility.

## 8.4 The response function

$g$  maps the normalised signal  $z_t = s_t / \hat{\sigma}_t^{(s)}$  to an exposure. Four canonical shapes:

| $g(z)$                  | Convexity | Turnover                  | Notes                                    |
|-------------------------|-----------|---------------------------|------------------------------------------|
| $z$ (linear)            | Highest   | Highest                   | Unbounded position on a signal outlier   |
| $\text{clip}(z, -c, c)$ | High      | High                      | The standard compromise; $c \approx 2$   |
| $\text{sign}(z)$        | None      | Lowest                    | Discards magnitude, and with it the skew |
| $z e^{-z^2/4} / 0.89$   | Moderate  | Lowest of the smooth ones | Fades extreme signals back toward zero   |

The last row is the response function popularised by Baz and co-authors (2015) and used in the deep-learning trend literature; the constant normalises the peak to 1. In my simulations it achieved a marginally higher Sharpe than the capped-linear response with roughly 30% lower turnover, which — after costs — is a real improvement.

Whether fading extreme signals is right is an empirical question with a surprising answer. Lempérière and co-authors (2014) find a clear saturation effect in the data: the return to a trend signal grows less than linearly and flattens for large signals, which they interpret as fundamentalist traders stepping in only once the mispricing is large enough to be worth fighting. That finding supports capping or bump-shaping, and argues against pure linearity. [Contested] — it is a single (interested-party) study, but the mechanism is plausible and the capping convention is near-universal in practice regardless.

The response function is also where you decide what your hit rate and skew will be (§5.7). That is worth deciding deliberately rather than discovering: a fund whose investors will redeem after two losing years in three should probably not run a pure linear response, whatever its Sharpe.

## 8.5 Position sizing

Assemble the pieces. For market  $i$  with signal  $z_{i,t}$ , price  $P_{i,t}$ , and contract point value  $V_i$ , the number of contracts is

$$n_{i,t} = \frac{\text{Capital} \times \sigma_{\text{portfolio}}^* \times \text{IDM} \times v_i \times g(z_{i,t})}{\hat{\sigma}_{i,t} \sqrt{A} P_{i,t} V_i \text{FX}_i}$$

where  $v_i$  is market  $i$ 's weight in the portfolio, IDM is the instrument diversification multiplier of §8.6,  $\hat{\sigma}_{i,t} \sqrt{A}$  annualises the volatility estimate, and  $\text{FX}_i$  converts the contract's currency to the book's. The structure is the point: every term except  $g(z_{i,t})$  is about risk, not about the view.

Two guardrails that belong here and are routinely omitted:

- A position cap per market, independent of the volatility scaling, so that a collapsed volatility estimate cannot produce an absurd position.
- A liquidity cap, as a fraction of average daily volume or open interest. This binds in exactly the markets that trend best.

## 8.6 Portfolio construction

The Sharpe multiplier of §5.9 is where the return comes from, so this section is rank 2 in the effort ordering.

The diversification multiplier. If you size each market to volatility  $\sigma^*$  and hold  $N$  of them with average pairwise strategy correlation  $\bar{\rho}$ , the portfolio's volatility is not  $\sigma^*$  — it is  $\sigma^* \sqrt{(1 + (N - 1)\bar{\rho})/N}$ , which is much lower. To hit your target you must scale up by

$$\text{IDM} = \sqrt{\frac{N}{1 + (N - 1)\bar{\rho}}},$$

the same factor as the Sharpe multiplier. For 50 markets at  $\bar{\rho} = 0.15$  that is 2.45 — you must run 2.45 times the naive gross exposure to achieve your target risk. Getting this wrong in either direction is a large error: too low and you run at half your intended risk for years; too high and a correlation spike delivers double.

Which is exactly the danger.  $\bar{\rho}$  is not stable. Trend systems converge to the same positions when many markets trend together, which is precisely when a shock hits several at once. [Fact] Realised correlations between trend-following programs, and between the same program's positions across markets, rise sharply in stress. The IDM computed from a calm-period correlation is therefore an over-estimate exactly when it is dangerous to over-estimate.

Three defences, in increasing order of intrusiveness:

1. Estimate  $\bar{\rho}$  on a long window and cap the IDM — commonly at 2.5. Crude and effective.
2. Use signed correlations. Baltas and Kosowski's (2013) refinement: what matters is not whether two markets are correlated but whether your positions in them are aligned. Two negatively correlated markets in which you hold opposite signs are a concentrated bet, not a diversified one. Compute the correlation of  $\text{sign}(z_i)r_i$  against  $\text{sign}(z_j)r_j$ .
3. Optimise directly on a shrunk covariance matrix, with a Ledoit–Wolf or factor-model estimator. Better in principle; in practice the estimation error at  $N \approx 50\text{--}100$  with a few years of data usually swamps the gain, and the resulting weights are unstable enough to add turnover.

[Practice] I would use (1) plus (2) and skip (3) unless  $N$  is large and the history is long.

Sector caps. Beyond correlations, impose structural limits: no more than some fraction of risk in bonds, in energy, in a single currency bloc. This is not elegant, but it protects against the case where the correlation estimate is simply wrong, which is the case that hurts.

Market selection. From §5.9, reducing  $\bar{\rho}$  from 0.15 to 0.05 is worth more than trebling  $N$ . In practice that means: prefer adding a market from an under-represented sector over adding a second contract in a well-covered one, and be sceptical of universes whose apparent breadth is many correlated instruments.

## 8.7 Trading rate, buffering, and the optimal-execution result

A naive implementation trades to its exact target every bar, which generates enormous and mostly pointless turnover — most of the change in the target is the volatility estimate's own noise.

The principled answer is Gârleanu and Pedersen (2013). With quadratic transaction costs and mean-reverting predictors, the optimal policy is:

Trade partially toward an “aim” portfolio, where the aim over-weights the more persistent signals.

$$\pi_t = (1 - \kappa) \pi_{t-1} + \kappa \text{aim}_t$$

The trading rate  $\kappa$  is increasing in risk aversion and decreasing in transaction costs — trade faster when risk matters more, slower when trading costs more. Two consequences that are easy to state and worth internalising:

1. You should never fully rebalance to your target. The optimal policy is a partial adjustment, always.
2. Signals with different decay rates deserve different weights in the aim than in the unconstrained optimum. A fast signal whose edge will be gone before you have finished trading into it should be down-weighted in the target, not just traded slowly.

[Practice] The practical approximation used in the industry is a no-trade buffer: compute the target position, and trade only if it differs from the current position by more than some fraction (commonly 10% of the average position, or half the position's own volatility). This captures most of the benefit of the optimal policy with none of the calibration burden. In my simulations a buffer of this size cut turnover by 30–50% with a Sharpe cost in the low single digits of percent.

## 8.8 Costs and capacity

Costs are not a haircut on trend-following; they are a binding constraint on its design. From my simulation, sweeping the kernel timescale and applying a proportional round-trip cost:

| EWMA span | Turnover | Gross SR | Net SR @ 1bp | @ 5bp | @ 20bp |
|-----------|----------|----------|--------------|-------|--------|
| 20        | 13.5×/yr | 0.20     | 0.16         | −0.02 | −0.68  |
| 80        | 7.1×/yr  | 0.32     | 0.29         | 0.20  | −0.14  |
| 320       | 3.9×/yr  | 0.37     | 0.36         | 0.31  | 0.12   |

Fast trend dies of costs; slow trend survives. At a 5 basis point round trip — realistic for liquid futures including market impact — a 20-span rule is already unprofitable while a 320-span rule keeps 84% of its gross Sharpe. This is the actual reason production systems run 1–12 month lookbacks, and it is a much better reason than “that is where the trends are”.

Three components to model:

- Spread and commission. Known, small in liquid futures (often under a basis point), and the easy part.
- Market impact. Scales as roughly the square root of participation (Tóth et al., 2011): trading  $Q$  against a daily volume  $V$  costs on the order of  $\sigma\sqrt{Q/V}$ . This is not linear, so doubling assets under management costs more than double.
- Slippage against the signal. Trend-followers trade in the direction the market has just moved, so they are systematically taking liquidity in the direction of price pressure. Cost estimates from symmetric samples underestimate what a trend program pays.

On capacity. The square-root law means capacity is finite and the constraint binds at the portfolio level: impact costs scale with  $\sqrt{\text{AUM}}$  while returns scale linearly, so net returns are concave and eventually decreasing in size. [Contested] Frazzini, Israel and Moskowitz (2018), using roughly 1.7 trillion dollars of live executions, find realised costs an order of magnitude below the academic proxies — which would push the capacity constraint far out. Their firm runs these strategies at scale, which is both why the data exists and why the result deserves a discount.

## 8.9 A reference implementation

The whole pipeline, in the order it must execute:

```

for each bar t:
    # --- 1. data, all using information available at t ---
    returns[i]    = stitched log return of market i          # §8.2
    sigma[i]      = blend(ewma_vol(returns[i], span=32),
                          ewma_vol(returns[i], span=180))    # §8.3
    sigma[i]      = max(sigma[i], vol_floor[i])

    # --- 2. signal ---
    fast[i]       = ewma(price[i], span=S)                   # §6.3
    slow[i]       = ewma(price[i], span=4*S)
    raw[i]        = (fast[i] - slow[i]) / (sigma[i] * price[i])
    z[i]          = raw[i] / ewma_sd(raw[i], span=2500)       # unit-scale signal
    g[i]          = clip(z[i], -2, +2)                        # §8.4

    # --- 3. sizing ---
    target[i]     = capital * vol_target * IDM * weight[i] * g[i]
                  / (sigma[i] * sqrt(A) * price[i] * point_value[i] * fx[i])
    target[i]     = clip(target[i], -pos_cap[i], +pos_cap[i]) # §8.5
    target[i]     = clip(target[i], -adv_cap[i], +adv_cap[i]) # liquidity

    # --- 4. portfolio overlay ---
    scale         = min(1, max_gross / sum(|target|))
    scale         *= drawdown_multiplier(equity_curve)        # §11.4
    target        = target * scale

    # --- 5. trade ---
    if |target[i] - position[i]| > buffer[i]:                  # §8.7
        trade to target[i]

```

Four things about this listing are load-bearing and are the ones people get wrong: every quantity is computed from data available at  $t$ ; the volatility estimate appears twice, once to normalise the signal and once to size the position, and they are conceptually different uses; the position caps are independent of the volatility scaling, so a broken  $\hat{\sigma}$  cannot produce an unbounded position; and the trade decision is a buffered comparison, not an assignment.

Signal parameters worth setting from §5.8 rather than from search:  $S \approx \tau/2$  where  $\tau$  is your estimate of trend persistence in bars, and a slow span of about  $4S$ .

## §8 Key takeaways

1. Effort ordering: data, volatility, universe, aggregation, timescale, costs, response, kernel shape. A mediocre indicator on clean data across 60 markets beats a superb one on dirty data across 5.
2. A futures price series is a construction, not an observation. Use return-stitching, roll on a schedule, and never let a rule see a back-adjusted level.
3. Blend a fast and a slow volatility estimate, floor it, and cap the position independently of it. Range estimators cut turnover by a third for free.
4. Compute every  $\hat{\sigma}_t$  causally. Full-sample volatility scaling is the most common invisible look-ahead in this literature.
5. Take direction from the slow filter and size from the volatility estimate. Dividing a slow signal's magnitude by a fast volatility estimate leaves noise in the ratio.
6. The diversification multiplier  $\sqrt{N/(1 + (N-1)\bar{\rho})}$  is both where the Sharpe comes from and the largest single sizing risk, because  $\bar{\rho}$  rises in stress. Cap it, and use signed correlations.
7. Never fully rebalance. Partial adjustment toward an aim portfolio is optimal under costs; a no-trade buffer captures most of it for free.
8. Costs determine the horizon. At 5bp round-trip, a 20-span rule is unprofitable and a 320-span rule keeps 84% of its gross Sharpe. This, not the location of “the trends”, is why production systems are slow.

## 9. When trends fade and fail

### 9.1 Three different claims

“Trend-following stops working” conflates three claims that have different evidence, different timescales, and different remedies.

| Claim                                                     | Timescale       | Remedy                                    |
|-----------------------------------------------------------|-----------------|-------------------------------------------|
| Within-trade decay — this particular trend is exhausting  | Days to weeks   | Exit and sizing rules; nothing structural |
| Regime dependence — trends are absent in this environment | Months to years | Risk reduction, diversification, patience |
| Secular decay — the premium itself has shrunk             | Decades         | Change the strategy or the business       |

Confusing the first with the third produces the classic failure mode of abandoning a strategy after a normal drawdown. Confusing the third with the first produces the opposite: patiently allocating to something that no longer works.

### 9.2 Within-trade: the sign of predictability flips with horizon

[Fact] Return autocorrelation is not one number; it is a function of lag whose sign changes. In broad terms, and with substantial variation by asset class:

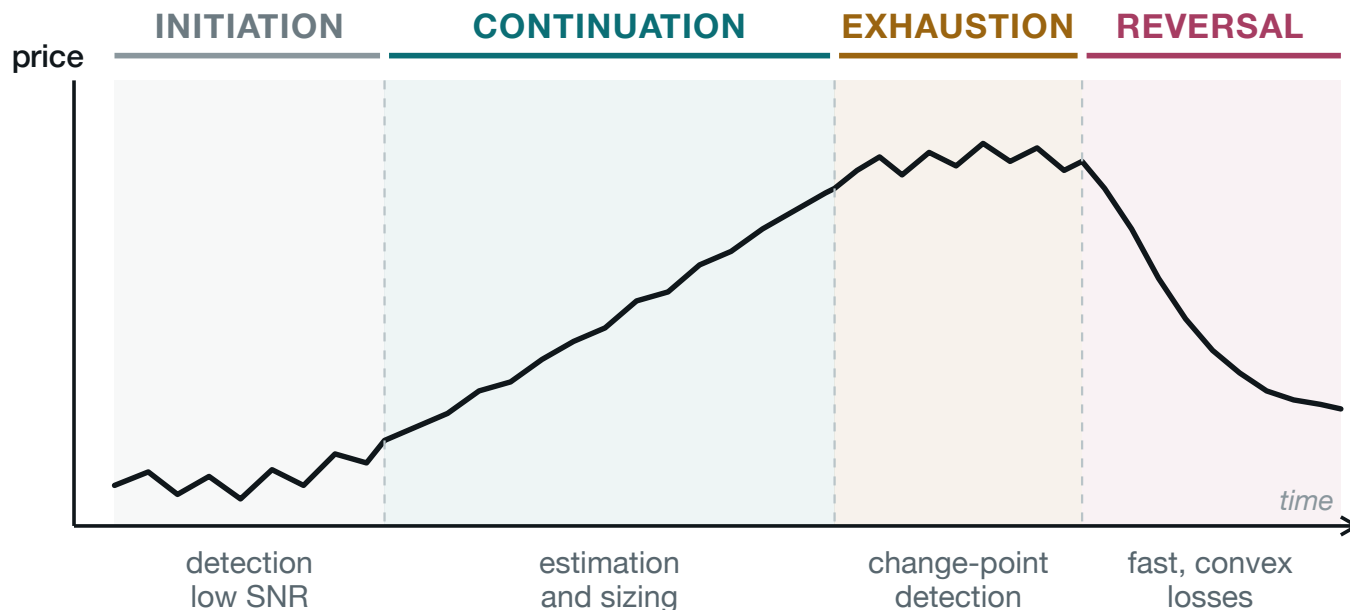
| Horizon            | Dominant sign                              | Mechanism                              | Consequence for a trend rule                    |
|--------------------|--------------------------------------------|----------------------------------------|-------------------------------------------------|
| Seconds to minutes | Positive in order flow, near zero in price | Order splitting                        | Not accessible to a daily system                |
| 1–5 days           | Negative (reversal)                        | Liquidity provision, bid-ask bounce    | Fast rules fight this; it is why they lose      |
| 1 month            | Negative in cross-section                  | Short-term reversal                    | The reason equity momentum skips the last month |
| 2–12 months        | Positive                                   | Underreaction, flow, macro persistence | Where trend-following lives                     |
| 1–3 years          | Weak                                       | —                                      | Nothing to harvest                              |
| 3–5 years          | Negative (reversal)                        | Overreaction correction, valuation     | Where a slow trend rule gets hurt               |

This is the structural reason a trend rule has an optimal horizon, and it is a stronger reason than the estimation argument of §5.8: even with perfect estimation, a rule whose kernel has mass at 3-day lags is betting on a negative autocorrelation with a positive sign.

The practical corollary appears in §7.2: check where your kernel’s mass actually is. A rule nominally “200-day” whose kernel peaks at lag 0 has a material fraction of its weight in the 1–5 day region, where the sign is wrong. This is a large part of why band-pass crossovers beat price-versus-average rules net of costs — they place almost no weight in the reversal zone.

### 9.3 The trend life cycle

[Practice / Hypothesis] The four-phase description below is an organising device with partial empirical support, not an established taxonomy. Its value is that it identifies which statistical problem you are solving at each stage.



| Phase        | Statistical problem                              | Signal quality                  | What to do                   |
|--------------|--------------------------------------------------|---------------------------------|------------------------------|
| Initiation   | Detection: is this the onset of drift or noise?  | Weak; most detections are false | Small size, wide tolerance   |
| Continuation | Estimation and sizing                            | Strongest                       | Full size, volatility-target |
| Exhaustion   | Change-point detection — the hardest of the four | Strong but decaying             | Reduce, tighten              |
| Reversal     | —                                                | Inverts abruptly                | Be flat or reversed          |

Initiation. Signal-to-noise is worst here by construction: you have the fewest observations of the new state. The lookback is the Type I / Type II dial, and no parameter resolves the trade-off — only your cost structure tells you where on the curve to sit (§8.8). Expect most detections to be false; §5.7's 25% hit rate is this fact.

Exhaustion. Candidate markers [Practice, weakly supported]: rising volatility with flat absolute price progress (a falling efficiency ratio, §6.9); negative acceleration while the trend estimate is still positive; divergence between price extremes and oscillator extremes; and crowding measures. None is a robust standalone signal in published evidence. Treat exhaustion detection as a reason to reduce risk, never as a reason to take the opposite position. The evidence does not support the reversal trade and the payoff structure punishes it.

## 9.4 Secular decay: has the premium shrunk?

This is the live question, and I will lay out both sides properly.

The case that it has decayed.

- Realised CTA index returns were poor from roughly 2009 to 2019 — a decade long enough that “normal drawdown” strains credulity.
- Assets in the strategy grew by an order of magnitude between 2000 and 2015. If the premium is compensation for providing liquidity to slow flows, more capital chasing it should compress it.
- The mechanism was published. Moskowitz, Ooi and Pedersen (2012) and its successors made the signal explicit, cheap products followed, and fees collapsed — a pattern documented for other anomalies after publication.
- Central-bank intervention from 2009 suppressed the volatility and rate dispersion that fixed-income trend-following feeds on. [Hypothesis]

The case that it has not.

- The 2009–2019 window is short relative to the statistical resolution of §5.9. Twenty years cannot distinguish Sharpe 0.4 from 0; a decade certainly cannot.
- 2022 was one of the best years in the strategy's history, driven by the bond and energy trends. A decayed premium does not do that.
- The long-history studies (§4.2) find the effect across eras with wildly different levels of participation, including eras with essentially no systematic capital at all.
- The most crowded thing about 2009–2019 was arguably not trend-following but the short-volatility complex, and the environment that hurt trend — persistently suppressed and rapidly mean-reverting volatility — is the environment that favoured its opposite.

My read. The premium is probably smaller than the pre-2000 sample suggests and probably not zero. Two specific components have almost certainly decayed: fast trend, which was competed away by execution technology and now sits inside the cost constraint of §8.8; and the fee-inclusive investor experience, which was compressed by competition — a good thing for investors and a bad thing for managers, and often mistaken for decay of the underlying premium.

## 9.5 Crowding and capacity

[Contested] Crowding is easier to assert than to measure. Three observable proxies:

1. Positioning data. Commitments-of-Traders reports classify managed-money positions. Crude, weekly, and heavily lagged, but directionally informative.
2. Return correlation among trend programs. If CTA returns become more correlated with each other and with a generic trend replicator, that is evidence of convergence on the same positions.
3. The signature of forced unwinds. Crowding shows up as conditional behaviour: sharp reversals in exactly the positions that a generic trend model would hold, at moments when the strategy is losing.

The mechanism to worry about is not that crowding lowers the average return — it does, mildly, through impact. It is that crowding changes the shape of the loss distribution: many participants holding the same positions with similar risk limits will deleverage simultaneously, which converts a moderate adverse move into a large one. That is a tail-risk story, and it is why §11.7 treats crowding as a risk-management topic rather than a return-forecasting one.

## 9.6 What would count as evidence

Because this question is so easily settled by whichever anecdote is nearest, it is worth writing down in advance what would change my mind.

Evidence the premium is gone:

- The variance-ratio profile of a broad futures panel converging to 1 at all horizons over a rolling 10-year window, tested on demeaned returns. This is the direct measurement, and by §5.4 it is the expected P&L.
- A statistically significant negative trend in per-decade trend-following Sharpe across the full 1880–present sample, robust to the choice of universe.
- Trend performance failing to recover in a period with large sustained macroeconomic dispersion — 2022 was the natural test, and it failed to fail.

Evidence it persists:

- Continued positive variance-ratio excess at 2–12 month horizons on demeaned returns in markets outside the original sample.
- Continued performance in markets with low systematic participation, where the crowding story does not apply.

[Practice] The variance-ratio monitor is cheap and I would run it as standing infrastructure: compute  $VR(q)$  on demeaned returns for  $q$  from 5 to 250 bars, per market and pooled, on a rolling window. It is a direct measurement of the thing you are being paid for, it is available long before P&L would tell you the same thing, and it requires no backtest.

## §9 Key takeaways

1. Distinguish within-trade decay, regime dependence, and secular decay. They have different timescales and opposite remedies, and conflating them causes the classic capitulate-at-the-bottom error.
2. Autocorrelation changes sign with horizon — negative at days, positive at 2–12 months, negative again at 3–5 years. A kernel with mass in the wrong zone bets against itself.
3. This is a stronger argument for the standard 1–12 month horizon than the estimation argument, and it reinforces the case for band-pass kernels.
4. Exhaustion detection is the hardest of the four life-cycle problems and no marker for it is well supported. Use it to reduce risk, never to reverse.
5. The secular-decay question is genuinely open. Fast trend has decayed; the fee-inclusive investor experience has compressed; the underlying premium probably persists at a reduced level.
6. A decade of poor performance is statistically uninformative at this Sharpe. Ten years cannot distinguish 0.4 from 0.
7. Crowding's main effect is on the shape of the loss distribution, not the mean — simultaneous deleveraging turns moderate moves into large ones.
8. Run a rolling variance-ratio monitor on demeaned returns. By §5.4 it measures your expected P&L directly, and it reacts long before your equity curve does.

## 10. Evaluation: testing a trend system honestly

### 10.1 The ladder

Test in stages, cheapest first, and stop at the first failure. Each stage answers a different question, and running them out of order wastes effort on systems that were never viable.

| Stage | Question                            | Test                                                            | Kill criterion                     |
|-------|-------------------------------------|-----------------------------------------------------------------|------------------------------------|
| 0     | Is there anything to harvest?       | $VR(q)$ profile on demeaned returns, pooled                     | No excess at any $q$               |
| 1     | Does the signal carry information?  | Correlation of $z_t$ with $r_{t+1}$ , pooled across markets     | Pooled IC indistinguishable from 0 |
| 2     | Does it survive the drift confound? | Repeat stage 1 on demeaned returns; alpha vs constant-long      | Effect disappears                  |
| 3     | Does it survive as a portfolio?     | Full backtest with sizing and aggregation                       | Sharpe below the alternative       |
| 4     | Does it survive costs?              | Re-run at $1\times$ , $3\times$ , $10\times$ your cost estimate | Sign flips within $3\times$        |

| Stage | Question                        | Test                                                    | Kill criterion                   |
|-------|---------------------------------|---------------------------------------------------------|----------------------------------|
| 5     | Does it survive the search?     | Deflated Sharpe;<br>walk-forward on untouched data      | Deflated Sharpe below 0          |
| 6     | Does it survive implementation? | Paper trade with real fills,<br>real rolls, real timing | Live/backtest gap<br>unexplained |

Stage 0 is unusual and is the payoff of §5.4: because the moving-average rule's expected P&L is half the variance-ratio excess, you can measure the opportunity before writing a backtest. If the pooled VR profile on demeaned returns is flat, there is nothing to find and no rule will find it.

Stage 4's kill criterion is deliberately harsh. Trend P&L is a small residual between two large terms (§5.5), so cost estimates that are off by a factor of two — which is normal — move the answer materially. A system that only works at your central cost estimate does not work.

## 10.2 Traps specific to trend

Beyond the usual backtesting hazards, these are the ones that bite trend systems in particular.

The drift confound (§5.3). Demean. This is the big one.

Full-sample volatility scaling.  $\hat{\sigma}_t$  computed on the whole sample leaks the future into every position. The resulting curve looks plausible, which is why this survives review.

Overlapping-window inference. A 12-month signal sampled daily produces observations that overlap by 251 days. Ordinary standard errors on such a series are too small by roughly  $\sqrt{L}$  — a factor of 16 for  $L = 252$ . Use Newey–West with a lag at least as long as the overlap, or sample non-overlapping, or block bootstrap.

Survivorship in the universe. Markets that stopped trading must be included.

Roll and timing look-ahead. Using the settlement price to decide a trade executed at that settlement; rolling based on same-day open interest; using a close that occurs after your decision time in another time zone.

Backtesting the smoother, not the filter. §6.7's look-ahead trap.

Cost model symmetry. Trend systems buy after the market has risen, so they trade with the flow. Costs estimated from a symmetric sample are optimistic.

Reporting trades rather than time. Trade-level statistics (§5.7) are seductive and the trade definition is arbitrary — position flips? sign changes? round turns? Report time-indexed returns; use trade statistics only for diagnostics.

## 10.3 Synthetic paths, where you know the answer

The most underused validation tool. Because the local level model of §5.8 has a known trend structure, you can generate paths with a specified  $\tau$  and  $\theta$  and check that your implementation recovers the known optimum.

What this catches that a backtest cannot:

- Implementation bugs that look like alpha. If your system beats the theoretical optimum on synthetic data with a known answer, you have look-ahead. This is the single most effective look-ahead detector I know, because it gives you a hard ceiling to compare against.
- Whether your estimator is even capable of extracting a known signal at your data length. If it cannot find a trend you know is there, no amount of real-data testing will help.
- Sizing errors. Feed the system a path with known volatility and check the realised risk matches the target.

The workflow: generate paths at your believed  $(\tau, \theta)$ , run the full pipeline, and compare against the Kalman-optimal benchmark. A well-built system should reach 80–95% of it. Substantially more means a bug; substantially less means the kernel or the sizing is wrong.

## 10.4 Bootstrapping a path-dependent strategy

Standard IID bootstrap is useless here: it destroys exactly the serial dependence the strategy exists to harvest, so it always says the strategy has no edge.

Two constructions that work:

- Stationary block bootstrap (Politis and Romano) with a mean block length comfortably longer than your kernel's centre of mass. Resampling blocks preserves within-block dependence. Use it to get honest confidence intervals on the Sharpe.
- The null bootstrap. Deliberately destroy serial dependence — shuffle returns within each market — and re-run the whole pipeline. This gives the distribution of your strategy's performance under the null of no predictability, with the drift, the volatility clustering (if you shuffle blocks of standardised returns and re-impose the volatility path), and your own selection procedure all intact. Your reported Sharpe must be compared to this distribution, not to zero. For a linear-response rule the null distribution is right-skewed (§1.5), so the naive comparison to zero is anti-conservative in a direction most people do not expect.

## 10.5 Multiple testing

You did not test one strategy. You tested  $S$  span choices,  $R$  response functions,  $V$  volatility estimators, and  $U$  universes, so the effective number of trials is large and the maximum Sharpe over them is biased upward. Bailey and López de Prado's deflated Sharpe ratio adjusts for the number of trials and the non-normality of returns; Harvey and Liu argue for haircutting reported  $t$ -statistics substantially.

Two things specific to trend research:

The trial count includes history. Trend rules were selected by the industry over fifty years on largely the same price series. Your independent trials are not the ones you ran; they are yours plus everyone's. This is the strongest form of the Sullivan–Timmermann–White critique and there is no statistical fix — only out-of-sample data, which means new markets and new eras.

Flat optima reduce the damage. §5.8's flatness result is good news here: if performance is insensitive to the parameter, then the maximum over a parameter sweep is only slightly above the average, so the selection bias from that sweep is small. Report the average over the sweep, not the maximum. For a flat objective that costs you almost nothing and removes most of the bias — a rare case where the honest choice is nearly free.

### §10 Key takeaways

1. Stage 0 is a variance-ratio profile, not a backtest. By §5.4 it measures the opportunity directly; a flat profile means no rule will work.
2. Demean every market's returns before any test. Undemeaned trend results are uninformative.
3. Overlapping windows inflate  $t$ -statistics by roughly  $\sqrt{L}$ . Correct or resample.
4. Test on synthetic paths with a known answer. Beating the Kalman-optimal benchmark is proof of look-ahead — the best such detector available.
5. Compare your Sharpe against a null bootstrap that preserves drift and volatility structure, not against zero. The null is right-skewed for linear-response rules.
6. Costs must be stress-tested at  $3\times$  and  $10\times$ . A system that works only at your central estimate does not work.
7. Report the average across your parameter sweep, not the maximum. With a flat optimum this costs almost nothing and removes most of the selection bias.
8. Your effective trial count includes fifty years of industry search over the same data. Only genuinely new markets and eras fix that.

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## 11. Pitfalls and risk management

### 11.1 The pitfall catalogue

Grouped by where they bite. The research-stage ones cost you time; the implementation ones cost you money; the live ones cost you the business.

| #              | Pitfall                              | Why it happens                                              | Detection                                       |
|----------------|--------------------------------------|-------------------------------------------------------------|-------------------------------------------------|
| Research stage |                                      |                                                             |                                                 |
| 1              | Drift confound                       | The test statistic is biased positive under the null (§5.3) | Demean; compare to constant-long                |
| 2              | Full-sample volatility scaling       | Feels like preprocessing, is look-ahead                     | Recompute causally; the curve should get worse  |
| 3              | Backtesting a two-sided smoother     | HP and $\ell_1$ fits use future data (§6.7)                 | Re-solve on expanding windows                   |
| 4              | Overlapping-window $t$ -statistics   | Inflated by $\sqrt{L}$                                      | Newey–West or block bootstrap                   |
| 5              | Universe survivorship                | Dead markets silently excluded                              | Count markets per year; it should not only rise |
| 6              | Optimising the lookback              | A flat, noisy objective with many trials (§5.8)             | Report the sweep average                        |
| 7              | Hidden negative kernel lobes         | Invisible in price space (§7.2)                             | Plot the return weights                         |
| Implementation |                                      |                                                             |                                                 |
| 8              | Roll gaps as returns                 | Unadjusted series inject fake returns                       | Check return distribution near roll dates       |
| 9              | Back-adjusted levels in a level rule | Breakout levels shifted by roll accumulation                | Never use adjusted levels; use returns          |
| 10             | Volatility estimate collapse         | Quiet market $\Rightarrow$ divide by near-zero              | Floor $\hat{\sigma}$ ; cap positions separately |
| 11             | IDM from calm-period correlations    | $\bar{\rho}$ rises in stress (§8.6)                         | Cap the IDM; use long windows                   |
| 12             | Symmetric cost estimates             | Trend trades with the flow                                  | Model impact asymmetrically                     |
| 13             | Trading to target every bar          | Most of the change is estimator noise                       | Buffer (§8.7)                                   |
| Live           |                                      |                                                             |                                                 |
| 14             | Gap risk in a synthetic option       | The convexity requires a tradeable market (§5.6)            | Stress-test with historical gaps                |
| 15             | Simultaneous deleveraging            | Everyone's risk limits bind at once (§9.5)                  | Monitor crowding; stagger limits                |
| 16             | Capitulating in a normal drawdown    | Ten years is statistically uninformative (§5.9)             | Pre-commit to a governance rule                 |
| 17             | Parameter drift                      | Quietly re-fitting after each drawdown                      | Version and date every parameter change         |

Number 17 deserves a sentence of its own, because it is the pitfall that survives everyone's process. A system that is re-tuned after each bad period is being fitted to the full sample, one drawdown at a time, and its live track record is an in-sample result. [Practice] Keep a dated log of every parameter change with its justification, and compute your track record on the parameters as they stood at each point in time. The difference between that and the current-parameter backtest is a direct measurement of how much you have overfitted.

## 11.2 Volatility targeting

Scaling positions by  $\sigma^* / \hat{\sigma}_t$  is the core risk mechanism, and it is worth being precise about what it does and does not achieve.

What it does. [Fact]

- Stabilises realised risk. This is its primary purpose and it works: in my simulation, the dispersion of realised P&L volatility across sub-periods fell by roughly three quarters when volatility scaling was applied.
- Improves the Sharpe ratio when volatility is persistent and negatively related to future returns. Harvey and co-authors (2018) document this across asset classes, with the effect strongest in equities — where the volatility–return relationship is strongest — and much weaker in commodities and currencies.
- Reduces the severity of the worst drawdowns, by deleveraging into rising volatility.

What it does not do.

- It does not help when volatility jumps rather than drifts. The estimator is backward-looking, so a discontinuous volatility increase is met at full size. Volatility targeting protects against the 2008 pattern (volatility rose over weeks) and not the February 2018 or March 2020 openings.
- It does not reduce risk in aggregate; it moves it. Deleveraging in high volatility means levering up in low volatility, and low-volatility periods are where fat-tailed jumps arrive from a low base.
- It can worsen skew. Selling into declines is what a volatility target mechanically does, and Harvey and co-authors find that while it improves Sharpe, its effect on skewness is unfavourable for some asset classes.

[Contested] And per §7.3, Kim, Tse and Wald (2016) argue much of the reported time-series momentum premium is attributable to the volatility scaling rather than the trend signal. If so, volatility targeting is not risk management bolted onto a strategy — it is a substantial part of the strategy, and should be evaluated as such.

### 11.3 Stops, and a result that surprises people

The folk wisdom is that stop losses are essential to trend-following. The theory says something more specific and more interesting.

Kaminski and Lo (2014) analyse when a stop-loss overlay improves the expected return of an underlying strategy. Their result:

A stop-loss rule adds expected return when returns are positively autocorrelated, and subtracts it when returns follow a random walk or are mean-reverting.

The intuition is direct. A stop sells after a loss. If losses tend to be followed by more losses — positive autocorrelation — selling is right. Under a random walk the future is independent of the loss, so the stop adds no information and subtracts the transaction cost plus the option value of the position. Under mean reversion the stop sells exactly what is about to recover.

Two consequences that matter:

1. A stop is not risk management; it is another trend signal. It is a fast, nonlinear, threshold-triggered trend rule stacked on top of your slow linear one. Evaluate it as a signal, with the same scrutiny, and check whether it is just a noisier duplicate of what you already have.
2. The case for stops in trend-following is therefore self-consistent but redundant. The same positive autocorrelation that justifies the stop already justifies the underlying rule, which will reduce the position anyway as the signal decays. Adding a stop mostly adds a faster kernel — with the cost profile of a fast kernel (§8.8).

[Practice] Where stops genuinely earn their place is for risks the signal cannot see: a hard per-position loss limit that catches data errors, model failures, and gap events. That is a different function from trend-following, and it should be sized and justified on those grounds — as an operational circuit-breaker, not as alpha. Set it far enough out that it never triggers in normal operation.

### 11.4 Drawdown control

The theoretical baseline is Grossman and Zhou (1993), who solve for the optimal investment policy subject to a constraint that wealth never falls below a fixed fraction of its running maximum. The solution scales exposure with the distance from the drawdown floor: as you approach it, you deleverage toward zero.

The practical form is a multiplier on the portfolio scale:

$$c_t = c_0 \cdot \psi\left(\frac{D_t}{D_{\max}}\right), \quad D_t = 1 - \frac{W_t}{\max_{s \leq t} W_s}$$

with  $\psi$  decreasing from 1 to 0 as current drawdown  $D_t$  approaches the tolerance  $D_{\max}$ . Three cautions:

- It converts a temporary loss into a permanent one. Deleveraging locks in the drawdown by removing the exposure that would recover it. Given §5.7's return distribution — where a handful of trades carry everything — being deleveraged during the recovery is expensive.
- It interacts badly with volatility targeting, since both deleverage in the same states, and the product of two multipliers can take exposure to almost nothing. Model them jointly.
- It is path-dependent in a way that defeats naive backtesting. The multiplier depends on your own equity curve, so a small change anywhere changes everything downstream. Confidence intervals from a block bootstrap (§10.4) are essential rather than optional here.

[Practice] I would use a gentle, slow-moving drawdown multiplier — a floor of perhaps 0.5 rather than 0, engaging only well into a serious drawdown — and I would treat it as an investor-retention and business-survival tool rather than as a return-improving one. It is not the latter, and the literature does not claim it is.

## 11.5 Tail and gap risk

The §5.6 point restated as a risk register: a trend-follower is long diffusive convexity and short gap risk. The convexity is manufactured by trading, and manufacturing requires a functioning market. The specific exposures:

- Limit moves. Commodity futures with daily limits can lock, preventing exit for consecutive sessions. Your position is frozen at exactly the moment the model wants it gone.
- Weekend and overnight gaps. The largest FX and rates moves have arrived when markets were closed — devaluations, peg breaks, referendum results, central bank surprises. The January 2015 Swiss franc de-peg moved roughly 30% in minutes; positions were sized on the prior volatility regime and there was no path through which to reduce them.
- Structural breaks in the price process itself. In April 2020 WTI futures settled at a negative price. Every log-return pipeline in existence produced either a NaN or a nonsense number. [Practice] Decide in advance what your system does with a non-positive price, and test it — the failure mode of an untested branch here is an unbounded position, not a missing data point.
- Correlated gaps. A single macro event gaps many markets at once, in the same direction as your positions, because your positions were all placed by the same recent trend.

Mitigations, in order of how much they actually help: position caps that bind independently of volatility scaling; sector and factor caps; explicit stress tests using historical gap events replayed against current positions; and — for programs of a size where it is affordable — actual purchased optionality on the largest exposures. Note that the last is the honest solution to a synthetic option's replication risk, and note also that it costs the premium you were trying to avoid paying.

## 11.6 Correlation risk

The IDM of §8.6 makes correlation a first-order risk rather than a modelling detail: if you have levered by  $2.45\times$  on the strength of  $\bar{\rho} = 0.15$  and correlations move to 0.6, your realised risk roughly doubles.

[Fact] Correlations rise in stress, and trend-following has an additional mechanism beyond the usual one: the strategy selects for correlated positions. A system holding 50 markets holds the ones that are trending, and markets trend together when a common macro factor is moving. Your effective  $N$  is smallest exactly when the shock arrives.

Monitoring, not forecasting. I would not try to forecast correlation. I would monitor three things continuously and act on them:

1. Realised portfolio volatility versus target. The direct measurement. If realised risk is running above target, the correlation assumption is already wrong.
2. Effective number of bets,  $N_{\text{eff}} = 1 / \sum_i \tilde{w}_i^2$  where  $\tilde{w}_i$  are the normalised principal-component variance shares of the current position vector. A collapse in  $N_{\text{eff}}$  is the crowding-inside- your-own-book signal.
3. Position concentration by sector and by macro factor — duration, dollar, equity beta, energy — since the trend signal does not know these exist.

## 11.7 Crowding and liquidity

Per §9.5, crowding's first-order effect is on the loss distribution rather than the mean. The mechanism is simultaneous deleveraging: many participants with similar positions and similar risk limits hit those limits at the same time, which turns a moderate adverse move into a large one.

Two defences that do not require you to measure crowding:

- Stagger your own risk limits so that your deleveraging is not a step function. If your volatility target, drawdown multiplier and position caps all bind at once, you have built a private version of the same problem.
- Trade slower than the crowd. The no-trade buffer of §8.7 has this useful side effect: you are not competing for liquidity in the same minutes as everyone whose signal flipped simultaneously.

And one that does: maintain a generic trend replicator — a simple, public, parameter-free version of the strategy — and monitor the correlation of your P&L to it. Rising correlation to the generic version means you are converging on the consensus position, which is the thing you would like to know before the unwind rather than during it.

## 11.8 The governance layer

The failure mode that ends trend-following programs is rarely a modelling error. It is the decision to stop, taken during a drawdown that was statistically unremarkable. §5.9's arithmetic is the whole argument: at a true Sharpe of 1.0, a three-year run with a negative Sharpe has a probability well above 5%, and at a true Sharpe of 0.4 it is close to a coin flip.

[Practice] Decide in advance, in writing, and before any money is at risk:

- What drawdown is consistent with the strategy working. Compute it from the simulated distribution, not from the historical maximum — the historical maximum is one draw and is almost certainly smaller than what you will eventually see.
- What evidence would constitute genuine deterioration (§9.6), stated as a measurement rather than a P&L threshold. The variance-ratio monitor is the right instrument because it is independent of your own returns.
- Who decides, on what schedule, with what information. A pre-committed review date removes the worst decisions, which are the ones made on the worst days.

This is not a soft topic appended to a technical document. Given a strategy whose statistical resolution is measured in decades and whose returns are right-skewed at the trade level, the governance rule is a risk parameter, and it is one of the few whose value you actually control.

### §11 Key takeaways

1. The research-stage pitfalls all reduce to the same thing: information from the future, or from the drift, entering the test. Demean, and compute every estimate causally.
  2. Log every parameter change with a date. A system re-tuned after each drawdown has an in-sample live track record.
  3. Volatility targeting stabilises realised risk [Fact] and improves Sharpe where volatility predicts returns; it does not protect against volatility that jumps, and it may be a larger part of the premium than the signal is.
  4. Stops add expected return only under positive autocorrelation (Kaminski and Lo). In a trend system they are a redundant fast signal. Keep a hard stop as an operational circuit-breaker, not as alpha.
  5. Drawdown control converts temporary losses into permanent ones and compounds with volatility targeting. Use it gently and justify it as business survival, not return.
  6. The convexity is synthetic and fails precisely at gaps. Limit moves, weekend jumps and negative prices are the exposures; decide in advance what the code does with each.
  7. Correlation is a first-order risk because the IDM levers on it, and trend-following selects for correlated positions. Monitor realised volatility and the effective number of bets rather than forecasting correlation.
  8. Pre-commit the governance rule. At these Sharpe ratios a multi-year losing run is unremarkable, and the decision to stop is the largest unhedged risk in the program.
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## 12. Synthesis

### 12.1 The framework in one page

Everything above reduces to three statements, each exact.

One. What it is. A trend-following system is a map from price history to position, increasing in recent returns:

$$\pi_t = \frac{\sigma^*}{\hat{\sigma}_t} g\left(\sum_k w_k \tilde{r}_{t-k}\right)$$

Kernel, response, risk scaling, aggregation. Every named indicator is a choice of the first two, and §7.2 gives the exact kernel for each.

Two. What it earns.

$$\mathbb{E}[\pi_t r_{t+1}] = \underbrace{\sum_k w_k \gamma_{k+1}}_{\text{autocovariance}} + \underbrace{\mu^2 \sum_k w_k}_{\text{drift}}$$

and for the canonical rule this is exactly  $\frac{1}{2}\gamma_0(\text{VR}(n) - 1)$ . Trend-following is long the variance ratio. The drift term is a confound to be removed, not a source of edge to be counted.

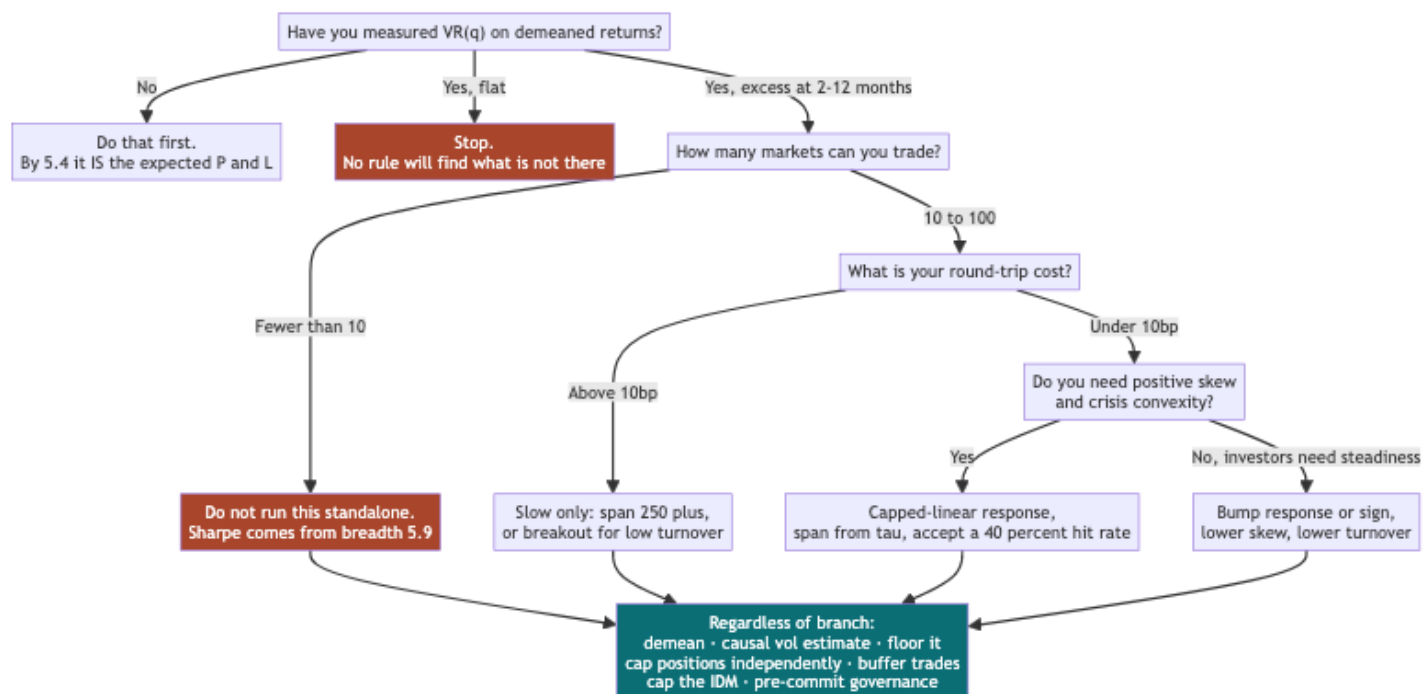
Three. What shape it is.

$$\sum_t x_t r_{t+1} = \underbrace{\frac{x_T^2 - x_0^2}{2\alpha(1-\alpha)}}_{\text{convexity}} + \underbrace{\frac{2-\alpha}{2(1-\alpha)} \sum_t x_t^2}_{\text{trend energy}} - \underbrace{\frac{\alpha}{2(1-\alpha)} \sum_t r_t^2}_{\text{premium paid}}$$

Long trend energy, short realised variance, in a ratio fixed by the filter speed. The convexity is real, it is paid for continuously, and it fails at gaps because it is synthetic.

Those three, plus the calibration of \$5.9 — a diversified Sharpe near 1.0 sits on a daily autocorrelation of about 0.005 — are sufficient to reason about almost any question in the field.

## 12.2 Decision tree



## 12.3 Building one from scratch

Staged, with a gate at each. Stages 1–3 are infrastructure and are not the interesting part; skipping them is the usual cause of failure.

| Stage            | Work                                                                     | Gate before proceeding                                                                                 |
|------------------|--------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|
| 1. Data          | Futures panel, return-stitched, dead markets included, scheduled rolls   | Return distribution shows no artefacts at roll dates; market count does not only rise                  |
| 2. Measurement   | $VR(q)$ profile on demeaned returns, per market and pooled               | A pooled excess exists at some horizon. If not, stop                                                   |
| 3. Risk plumbing | Causal volatility estimates, position sizing, realised-risk monitor      | On synthetic paths of known volatility, realised risk matches target within a few percent              |
| 4. Signal        | One EWMA crossover, span from $\tau$ , capped-linear response            | On synthetic paths, performance is 80–95% of the Kalman-optimal benchmark. Above that means look-ahead |
| 5. Portfolio     | IDM, sector caps, signed correlations                                    | Realised portfolio volatility tracks target out of sample                                              |
| 6. Costs         | Impact model, no-trade buffer, turnover budget                           | Sign survives $3 \times$ your cost estimate                                                            |
| 7. Validation    | Block bootstrap, null bootstrap, deflated Sharpe, walk-forward           | Sharpe exceeds the null distribution, not zero                                                         |
| 8. Risk          | Gap stress tests, drawdown multiplier, crowding monitor, governance rule | Written, dated, and agreed before capital                                                              |

## 12.4 Ten things I would tell someone starting today

1. Measure the variance ratio before you write a backtest. It is the expected P&L (\$5.4), it takes an afternoon, and it will tell you whether to continue.
2. Demean. Every market, every test, always. This one line removes the most common way trend research fools its author.
3. Spend your effort on the volatility estimate and the correlation structure, not on the indicator. That is where the return is (\$7.3).

4. Set the lookback from trend persistence, not from backtest P&L. Span  $\approx 2\tau$ . The optimum is flat, so approximately right is right enough, and optimising it is how you overfit.
5. Trade slowly. Costs, not signal quality, set the horizon (§8.8), and partial adjustment beats full rebalancing always (§8.7).
6. Diversify across genuinely different markets. Lowering  $\bar{\rho}$  from 0.15 to 0.05 beats trebling the market count.
7. Expect to be wrong three times out of four. The hit rate is a design choice, not a quality measure, and the strategy's entire profit sits in a handful of trades (§5.7).
8. Test on synthetic data where you know the answer. Beating the theoretical optimum is proof of a bug, and it is the best look-ahead detector there is.
9. Cap positions independently of the volatility scaling. A collapsed  $\hat{\sigma}$  should produce a capped position, not an unbounded one.
10. Write the governance rule before you have money at risk. At this Sharpe, a three-year losing run is unremarkable, and the decision to stop during one is the largest risk in the program.

## 12.5 What is and is not known

Established. The algebra: the P&L identities of §5.2, §5.4 and §5.5 are theorems and are not in dispute. The kernel equivalences of §7.2 are exact. The statistical calibration of §5.9 — that the effect is individually undetectable and only visible in aggregate — follows from arithmetic. That volatility scaling substantially improves risk-adjusted performance is replicated. That the payoff is convex in the underlying's move, and that hit rate is a property of the response function, is derivable and has been derived.

Contested, with the disagreement worth respecting on both sides. Whether the time-series momentum premium survives the drift correction (Huang and co-authors versus Moskowitz and co-authors). How much of the premium is really the volatility scaling (Kim, Tse and Wald). Whether the premium has decayed since 2009. What CTA investors have actually earned net of fees and biases, where the critique (Bhardwaj, Gorton and Rouwenhorst) is stronger than the defence.

Not known. Why trends exist. Five mechanism families have support and none is decisive; the cross-section of results — trend-following working well in the most information-efficient markets in the world — is not comfortably explained by any of them. Nor is there a reliable way to detect trend exhaustion, or to measure crowding in real time, or to forecast when the strategy will work.

What I would flag as the largest open risk in the field, and it is a methodological one: essentially all of the long-history evidence that trend-following persists comes from firms that sell it. That is not a reason to dismiss the evidence — they are the only groups with the data, and two of them with different books reached the same answer independently. But it does mean the single most valuable contribution someone could make to this literature is a long-history study, on demeaned returns, by an author with no position.

Trend-following remains, to me, the most intellectually honest strategy in systematic trading: its mechanism is a measurable statistic, its payoff shape is an algebraic identity, and its failure modes are known and enumerable. Whether it still pays enough to be worth running is a separate question, and one the variance-ratio monitor of §9.6 will answer for you long before your P&L does.